

UNIVERSITY OF THE PUNJAB



SOFTWARE DESIGN DESCRIPTION

For

HotelHype

“Find The Perfect Stay, Your Way”

Version 1.0

Project ID: 20-KS-BSIT-09

By

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Bachelor of Science in INFORMATION TECHNOLOGY (2020-2024)

DECLARATION

The work reported in this project was carried under the supervision of Project Supervisor, **Sir Hasan Raza**, at Govt. Graduate College, Civil Lines, Sheikhpura.

We hereby declare that this project and the contents of the project are the product of our research and no part has been copied from any other written or published source (except the references, standard mathematical or genetics models/equation/formulas/protocol, etc.).

We further declare that this work has not been submitted for the award of any other degree diploma. The university may take action if the provided information is found inaccurate at any stage.

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ACKNOWLEDGEMENTS

In the name of **ALMIGHTY ALLAH**, the most merciful, the most beneficent, Who is the Lord and Owner of all authorities and capabilities Who endowed His greatness on us. He is the entire source of knowledge of wisdom and Who blessed us with the ability to do work. We all grateful to the **Prophet Muhammad (PBUH)** who gave us the spirit to learn. No doubt in it because of **ALMIGHTY ALLAH** that today we are able to complete our project.

We are also thankful to our honorable Supervisor **Sir Hasan Raza**, Department of Computer Science of Govt. Graduate College, Civil Lines, Sheikhpura for his guidance, encouragement, support, and admirable help. He was always encouraging to come to the project work. In fact, without his efforts and guidance, it might not have been possible to complete the work.

We would like to thanks all the faculty of Information Technology of Govt. Graduate College, Civil Lines, Sheikhpura for this great support and help for the proper completion of our project work. We should feel it necessary to express our exclusive love and adoration in our parents and other family members for their support, encouragement, and tremendous contribution to our project completion.

May all they live, long to see our dreams being fulfilled. **Ameen!**

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PROJECT INFORMATION

Project Title	HotelHype
Objective	A comprehensive hotel recommendation system that allows users to search for hotels based on their preferences, while integrating sentiment analysis and OTP-based user verification to ensure secure and personalized recommendations.
Undertaken by	Amina Atiq, Ayesha
Supervised by	Sir Hasan Raza
Starting Date	15/3/2024
Completion Date	15/8/2024
Tools Used	Visual Studio Code, Jupyter Notebook, MYSQL Workbench

PREFACE

Thanks for your interest in this project report. This is a final year project report written by students of **Bachelor of Science in Information Technology** at the **University of the Punjab**. I am **Amina Arshad** and my project partner **Ayesha** studying in BS (Hons) of **Govt. Graduate College, Civil Lines, Sheikhpura**.

This is a technical report, we'll explain the idea of our project and the implementation process in detail and systematically. To enable you to understand this project more clearly. This report is divided into 20 topics; each topic with a specific focus. We hope this approach makes it easy for you to read our project report easily. Lastly, if you have any comment you are Welcome, feel free to contact us at:

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ABSTRACT

Hotel Hype is the ultimate web-based destination to revolutionize the hotel search experience for users. This system utilizes real-time data scraping of hotels, followed by sentiment analysis to evaluate user reviews and assess hotel quality to provide users with valuable insights and optimize hotel selection. Users can get hotel recommendations by selecting their desired cities, facilities, and prices. A live location features, users can find nearby recommended hotels. Users can also write the review. This is especially useful for travelers looking for last-minute help or exploring local options. This system provides a wonderful way to discover the best hotels according to the customer's preferences. Analyzing reviews makes the platform a rich collection of users experience data. Hotel owners and partners (called "Hype Partners") can list their hotels, manage their listings, update information, and cancel hotels, if necessary, by registering to our hotel hype listing form. Passwords during registration are encrypted to increase the user security. The hotel hype website includes admin panel that allows administrator to approve or reject hotel listings, manage user accounts, and ensure accuracy and quality of the web content. It also provide partner panel to the listed partner of the hote

1. INTRODUCTION

Hotel Hype is an innovative website. This website enhances the travel experience by providing hotel reviews and personalized recommendations. This website simplifies the process of finding and reviewing the best hotels by combining real-time data, user feedback and reviews sentiment analytics. This website differentiates itself by maintaining a rich and up-to-date hotel information database using user reviews and real-time web scraping.

Users can search their preferences and hotels are recommended based on preferred city, facilities, and prices. Users can also avail live location feature to get instant recommendation results according to their current location. Hotel owners can list their hotels, manage listing, update hotel information, cancel listing any time and do account setting. And the Administrators on their panel have access to approve or reject hotel listing, manage user accounts, manage hotel listing, and update hotels data. The hotel hype interface provides a genuine and secure platform that allows travelers to make informed decisions making and improve their entire hotel experience.

1.1 Scope

Hotel Hype is a website designed to simplify and enhance the travelers experience for hotel discovery and explore top accommodations. This site features a personalized hotel recommendation system that allows user to search and find hotels based on their location, facilities, and budget preferences. Users can get the live location feature for quick recommendations and register on the site; submit reviews of the hotels they have visited. This site uses confidential information and adheres to privacy standards to protect user security and privacy. Hotel Hype provides an advanced dashboard to effectively list, update and manage their hotels. Hotel Hype aims to provide efficient and effective services to the travelers and hoteliers, making it a versatile and comprehensive platform for hotel reviews and recommendations.

1.2 Purpose

The main purpose of Hotel Hype website to provide a user-friendly platform that facilitates travelers for hotel discovery and accommodations using user-reviews and data analytics to help them make informed decisions while choosing a hotel by providing personalized recommendation based in their specific preferences. This site provides effective management tools for hotel owners and managers and interact with customer's feedback. Hotel Hype following strict data protection standard to provide the user data privacy and security by using encryption password storage. Overall, the main objective of the website is to provide secure and reliable experience for travelers and hoteliers with the aim of improving overall hotel industry.

2. PROBLEM STATEMENT

Travelers today are faces significant challenges by the vast number of inconsistent reviews, complexity of filtering opinions through biased rating score of hotels. Furthermore, the process of searching hotels according to user preferences. Hotel owners face difficulties in managing the online status of hotels, including the tools for hotel listing, hotel information updates, account setting, cancellation of hotel listing, response to feedback, customer engagement and much more. Data security is the major concern for users and hotel owners. Traditional platforms are influenced by individual biases, leading to the poor decision making and dissatisfaction among the travelers. To overcome all these challenges, Hotel Hype website is aim to offering reliable and user-friendly platform. This site overcome the issues of biased hotel ratings by doing sentiment analysis of reviews to get the accurate and trustworthy hotel rating score, building trust over customers. To simplify the searching process, this website provides live location to get quick recommended hotels. This website also ensures the security of users data by password encryption storage to provide reliable experience to users

2.1 Overall Description

Overall Description of our website is that it is providing personalized hotel recommendations based on user preferences such as location, facilities and price. It uses sentiment analysis to provide real and unbiased hotel ratings on the user reviews. A live location features also enhance the users quick searching for hotels. For hoteliers, Hotel Hype offers a management panel to efficiently handle hotel listing and customer feedback. Overall, Hotel Hype aims to deliver reliable, trustworthy and intuitive solutions for both travelers and hoteliers, enriching the overall experience in the hospitality industry.

2.2 Product Perspective

Hotel Hype Website should provide user friendly interface and exciting advance features to the users, hotel hype partner to make this platform more seamless, reliable and secure. There will be a user-friendly GUI. Hotel Hype includes an administration panel to help manage listings and guest feedback for strong online engagement. The platform also prioritizes security by ensuring data privacy and strong data protection. From a broader product perspective, Hotel Hype aims to build trust and value for travelers and hoteliers by improving their overall experience in the hotel industry.

2.3 Operating Environment

The website will require following operating environment:

- Web-based application can be accessed from any device.
- Minimum 4 GB Ram for smooth performance.
- A dual-core processor.
- 1366x768 pixels or higher for optimal user experience.
- Internet connection.

2.4 Design and Implementation Constraints

Hotel Hype is a user-friendly web-based application running over all devices smoothly like PC's and mobile android application running over android cell phone. The application is developed using Front-end technologies (Html, CSS, JS), back-end technologies (Python, Jupyter notebook (for Sentiment analysis)) and flask is used to build backend of the website. It uses as a web framework which handles HTTP request and process user's input. It is also used for session management and routing for smooth navigation through website. The interface is in the form of Website. Html, CSS and JS is used for structuring the content of web, styling of web content and enabling dynamic features and handling user inputs. For the development of backend of this website we firstly scrap hotel reviews data from google maps by making our own python scraper, then do data analysis (data cleaning, data processing and transformation) through python. For sentiment analysis we use pre-trained ML library for labeling data, then find the rating score from the review's sentiment analysis. After that make rule-based recommendation model for handling the searching according to user preferences. We create database (MYSQL) that is relational database. We create login, registration activity for user, admin panel activity for administrators to control all activities, Hype partner panel functionalities for controlling the hotel listing, hotel information updating, canceling any listed hotels. Write a review activity and live location features for users.

3. REQUIREMENT IDENTIFYING TECHNIQUES

3.1 Admin's Scenarios:

List of possible use cases of **Admin**:

- UC_Login
- UC_DeleteUserAccounts
- UC_AddNewHotel
- UC_DeleteHotel
- UC_ManageRequests
- UC_Logout

Use Case Diagram (Admin)

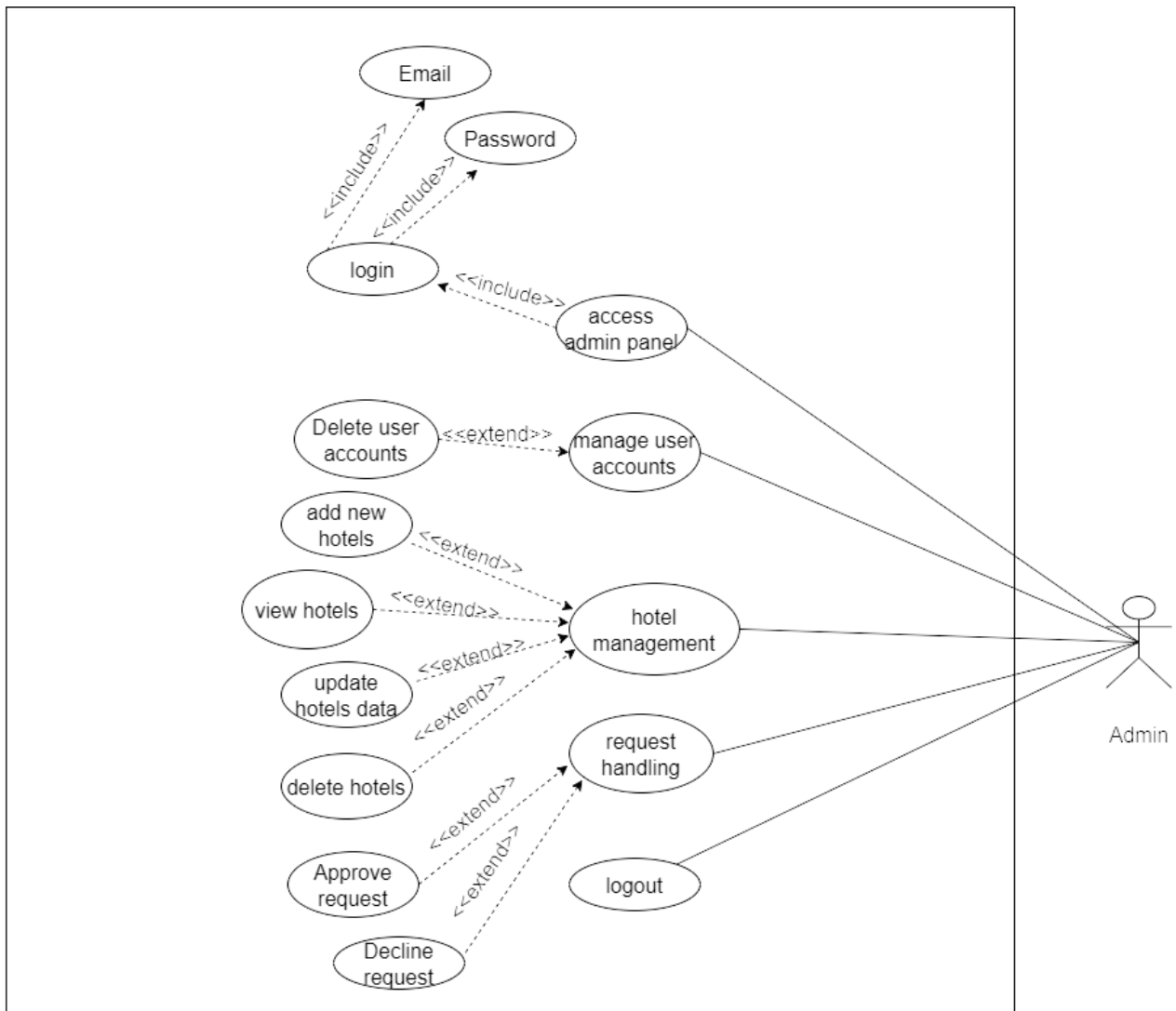


Figure 1: Use Case Diagram of Admin

Use Case Discussed

Below is the use case descriptions for all of the use cases mentioned above in user's scenarios.
(r, n.d.)

3.1.1 UC_Login

Use Case ID:	UC_01
Primary Actor	Admin
Use Case Name:	UC_Registration
Description:	In this use case, the admin logs into the system to access the admin panel and manage the backend functionalities of the application.
Preconditions:	The admin has a valid username and password.
Basic Flows:	• Admin navigates to the login page. • Admin enters their username. • Admin enters their password. • Admin submits the login form. • System verifies the credentials. • On successful verification, the admin is redirected to the admin panel.
Exceptional Flows:	• If the username is not recognized, the system displays an error message indicating that the username does not exist. • If the password is incorrect, the system displays an error message indicating that the password is incorrect.
Postconditions:	The admin is logged into the system and has access to the admin panel where they can manage users, hotels, and requests.

3.1.2 UC_AddNewHotels

Use Case ID:	UC_02
Primary Actor	Admin
Use Case Name:	UC_AddNewHotels
Description:	In this use case, the admin adds new hotel details through admin panel. The details are then stored in the database as a new hotel entry.
Preconditions:	The admin is logged into the admin panel and has access to the hotel management section.
Basic Flows:	• Admin accesses the admin panel. • Admin navigates to the "Add Hotel" section. • Admin clicks on the "Add Hotel" button. • Admin fills in the required hotel details (e.g., hotel name, location, description, facilities, price). • Admin clicks "OK" to submit the details. • The system stores the new hotel details in the database. • The system confirms the addition of the new hotel.
Exceptional Flows:	• If any required fields are missing, the system displays an error message indicating which fields need to be completed. • If the entered details do not meet validation criteria (e.g., invalid price format), the system displays an error message.
Postconditions:	The new hotel details are successfully stored in the database, and the hotel is available for management and display within the system.

3.1.3 UC_DeleteHotel

Use Case ID:	UC_03
Primary Actor	Admin
Use Case Name:	UC_DeleteHotels
Description:	In this use case, the admin deletes a specific hotel from the system using the admin panel.
Preconditions:	The admin is logged into the admin panel and has access to the hotel management section.
Basic Flows:	• Admin navigates to the hotel management section of the admin panel. • Admin views a list of all hotels with their details. • Admin locates the specific hotel to be deleted. • Admin clicks on the "Delete" button next to the hotel. • The system deletes the hotel from the database. • The system displays a confirmation message indicating that the hotel has been successfully deleted. • If the deleted hotel is of some listed partner then a email notification will be send to him by system that “your listed hotel is deleted from website by admin
Exceptional Flows:	• If the deletion process encounters an error (e.g., database connectivity issue), the system displays an error message and the hotel remains listed.
Postconditions:	The specified hotel is removed from the database, and a confirmation message is shown to the admin. The hotel no longer appears in the hotel management section and on the website.

3.1.4 UC_DeleteUser

Use Case ID:	UC_04
Primary Actor	Admin
Use Case Name:	UC_DeleteUser
Description:	In this use case, the admin deletes a user account from the system through the User Management section of the admin panel.
Preconditions:	The admin is logged into the admin panel and has access to the User Management section.
Basic Flows:	• Admin navigates to the User Management section in the admin panel. • Admin views a list of users with their details and a delete button for each user. • Admin selects the user to be deleted by clicking the corresponding delete button. • System deletes the selected user from the database. • System displays a confirmation message indicating that the user has been successfully deleted. • If the account is of some listed partner then a email notification will be send to him by system that “your account is removed from website by admin”
Exceptional Flows:	• If there is a failure in deleting the user (e.g., database error), the system displays an error message indicating that the deletion was unsuccessful.
Postconditions:	The selected user account is removed from the system, and the admin is notified of the successful deletion. The User Management section is updated to reflect the changes.

3.1.5 UC_RequestHandling

Use Case ID:	UC_05
Primary Actor	Admin
Use Case Name:	UC_RequestHandling
Description:	In this use case, the admin manages hotel listing requests. The admin reviews requests in the request management section, approves or denies them, and the system sends appropriate emails to the users based on the action taken.
Preconditions:	The admin is logged into the system and has access to the request management section. There are pending hotel listing requests in the system.
Basic Flows:	• Admin navigates to the request management section in the admin panel. • Admin views the list of pending hotel listing requests. • Admin decides to either approve or deny the request. • If approved: a. The system updates the database to list the hotel. b. The system sends an email to the user notifying them that “Your hotel has been approved for listing” • If denied: a. The system does not add the hotel to the database. b. The system sends an email to the user notifying them that “Your hotel has been denied for listing.”
Exceptional Flows:	• If the system encounters an error while updating the database, an error message is displayed, and the email is not sent.
Postconditions:	The hotel listing request is either approved or denied, and the appropriate action is taken in the database. The user receives an email notification about the status of their hotel listing request.

3.1.6 UC_Logout

Use Case ID:	UC_06
Primary Actor	Admin
Use Case Name:	UC_Logout
Description:	In this use case, the admin logs out of the system to end their current session and ensure that their access to the admin panel is terminated.
Preconditions:	The admin is currently logged into the system and has access to the admin panel.
Basic Flows:	• Admin navigates to the admin panel. • Admin clicks on the "Logout" button. • System terminates the admin's session. • Admin is redirected to the home page.
Exceptional Flows:	• If the logout process fails (e.g., due to a system error), the system displays an error message indicating that the logout was unsuccessful.
Postconditions:	The admin is logged out of the system, and their session is terminated. They are redirected to the home page, and their access to the admin panel is no longer available until they log in again.

3.2 Hotel Partner's Scenarios:

List of possible use cases of **Hotel Listed Partner**:

- UC_Login
- UC_ManageHotel
- UC_UpdateAccountDetails
- UC_ListHotel

Use Case Diagram (Partner)

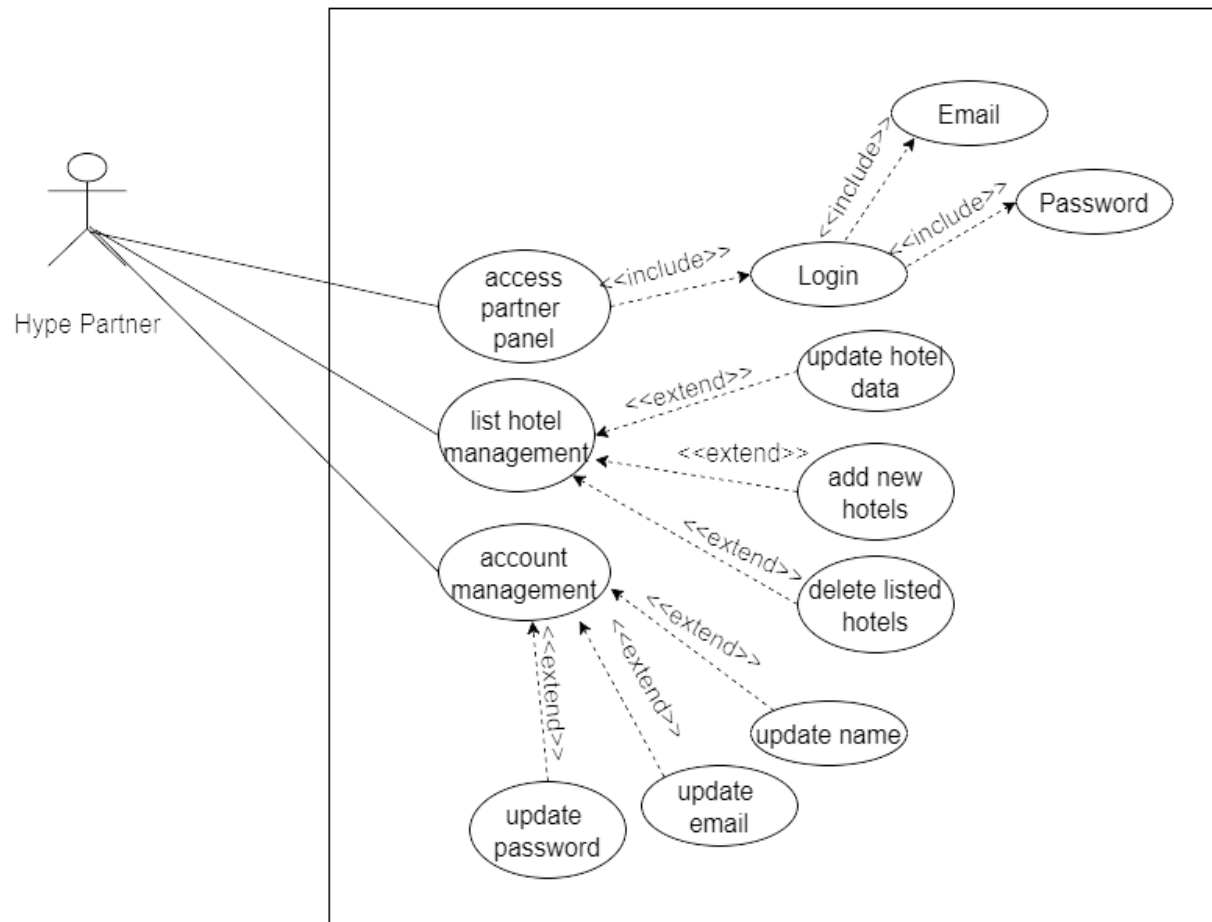


Figure 2: Use Case Diagram of Listed Hotel Partner

Use Case Discussed

Below is the use case descriptions for all of the use cases mentioned above in listed Partner's scenarios.

(r, n.d.)

3.2.1 UC_Login

Use Case ID:	UC_01
Primary Actor	Hotel Partner
Use Case Name:	UC_Login
Description:	This use case allows a partner to log in to the system using email and password. The system verifies the credentials and grants access to the partner panel if they are a valid listed partner. If the credentials are incorrect or the partner is not listed, an error message is displayed.
Preconditions:	The partner must be a registered user with valid credentials.
Basic Flows:	• Partner navigates to the dashboard option. • Partner enters email and password. • Partner submits login request. • System verifies credentials. • If valid, system grants access to the partner panel. • If invalid, system displays an error message.
Exceptional Flows:	• If email is not registered in the system, display "You are not a listed Partner." • If the email and password do not match any registered partner, display "Invalid login credentials."
Postconditions:	The partner is either granted access to the partner panel or presented with an error message based on the validity of the credentials.

3.2.2 UC_ManageHotels

Use Case ID:	UC_02
Primary Actor	Hotel Partner
Use Case Name:	UC_ManageHotels
Description:	This use case allows a partner to manage their listed hotels through the partner panel. The partner can update or delete existing hotel listings. If a partner deletes all of their listed hotels, they will no longer be a listed partner on the website.
Preconditions:	The partner must be logged in and have at least one hotel listed.
Basic Flows:	• Partner navigates to the partner panel. • Partner selects a hotel from the list of their listed hotels. • Partner chooses to update or delete the selected hotel. • For update: Partner makes changes to the hotel details and details will be updated in database. • For delete: Partner confirms the deletion of the hotel. • System processes the update or deletion. • If the partner deletes all listed hotels, the system removes the partner's listing status.
Exceptional Flows:	• If the update request contains invalid data, an error message is displayed and the partner is prompted to correct the data. • If the partner tries to delete all listed hotels but encounters an issue, an error message is displayed.
Postconditions:	• The hotel details are updated or the hotel is removed from the partner's list. • If all hotels are deleted, the partner is no longer listed on the website.

3.2.3 UC_UpdateAccountDetails

Use Case ID:	UC_03
Primary Actor	Hotel Partner
Use Case Name:	UC_UpdateAccountDetails
Description:	This use case allows a partner to update their account details such as email, password, and username from the partner panel. The system validates the input details and applies the changes if they are valid.
Preconditions:	The partner must be logged into the partner panel with valid credentials.
Basic Flows:	• Partner navigates to the account settings section in the partner panel. • Partner selects the option to update account details. • Partner inputs new email, password, or username. • Partner submits the update request. • System validates the new details. • If valid, the system updates the account details and confirms the changes. • If invalid, the system displays an error message explaining the issue.
Exceptional Flows:	• If the new email is already in use, display "Email is already registered." • If the new password does not meet security requirements, display "Password invalid." • If the new email format is invalid, display "Invalid email format."
Postconditions:	• The partner's account details are updated if all input details are valid. • The partner is notified of successful updates or any issues encountered during the process.

3.2.4 UC_ListHotel

Use Case ID:	UC_04
Primary Actor	Hotel Partner
Use Case Name:	UC_ListHotel
Description:	This use case allows a listed partner to add a new hotel to the system. The partner fills out the required hotel details, and submits the information. The listing request will be sent to the admin and system send email notification to admin for listing request.
Preconditions:	The partner must be logged in and have access to the partner panel.
Basic Flows:	• Partner navigates to the 'List Hotel' section in the partner panel. • Partner fills in all required hotel details including hotel name, location, description and other relevant information. • Partner submits the hotel listing form. • System confirms the submission and provides a notification that the listing request has been sent to the admin for approval. • System will send email notification to admin for listing request.
Exceptional Flows:	• If required fields are missing or invalid, the system displays an error message indicating the issue ("Please fill in all required fields").
Postconditions:	• The Listed hotel request is sent to admin for review . • The partner receives a confirmation message that the listing request has been sent to the admin.

3.3 Visitors's Scenarios:

List of possible use cases of **Visitor**:

- UC_Registration
- UC_EmailVerificationOTP
- UC_BrowseSearch
- UC_ViewHotelDetails
- UC_LiveLocationSearch
- UC_ListHotel
- UC_ReviewSubmission

Use Case Diagram (Visitor)

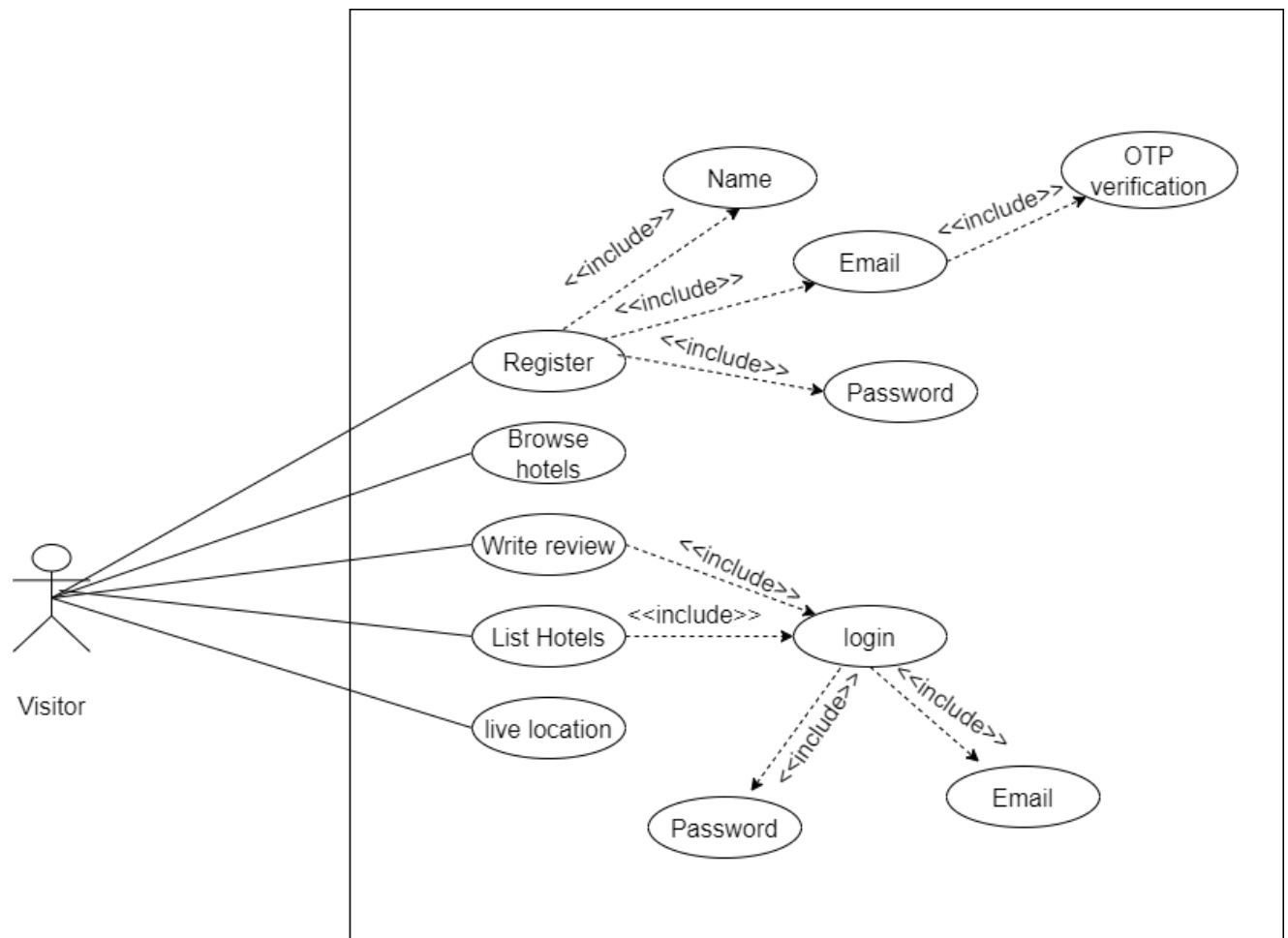


Figure 3: Use Case Diagram of Visitor

Use Case Discussed

Below is the use case descriptions for all of the use cases mentioned above in Visitor's scenarios.
(r, n.d.)

3.3.1 UC_Registration

Use Case ID:	UC_01
Primary Actor	Visitor
Use Case Name:	UC_Registration
Description:	In this use case, a visitor registers to access the system by providing their email and password. The system verifies the provided details and registers the visitor through OTP verification if all validations are successful
Preconditions:	None
Basic Flow:	• Visitor navigates to the registration page. • Visitor enters their email and password. • Visitor submits the registration form. • System validates the email and password. • If the email and password are valid, the system registers the visitor and sends a confirmation email or OTP for verification.
Exceptional Flows:	• If the email is already registered, the system displays a message: "This email is already registered." • If the email format is invalid, the system displays a message: "Invalid email format."
Postconditions:	After successful registration and OTP verification, the visitor is registered and can log in to access the system. If registration fails due to validation issues, the visitor is prompted to correct the information and resubmit.

3.3.2 UC_EmailVerificationOTP

Use Case ID:	UC_02
Primary Actor	Visitor
Use Case Name:	UC_EmailVerificationOTP
Description:	In this use case, after the user registers, an OTP (One-Time Password) is sent to their email for verification. The user must enter this OTP to complete the registration process.
Preconditions:	The user has completed the registration process by submitting their email and other details. An OTP is generated and sent to the user's email.
Basic Flow:	• User submits the registration form with their email and other details. • System generates an OTP and sends it to the user's email. • User enters the OTP received via email. • User submits the OTP. • System verifies the OTP against the one generated. • If OTP is valid, the system completes the registration process and confirms the user's registration. • User is notified of successful registration.
Exceptional Flows:	• OTP entered by the user is incorrect or expired. • System displays an error message indicating that the “OTP is invalid or expired”
Postconditions:	After successful OTP verification, the user is fully registered and can log in to their account. If OTP verification fails, the user is prompted to retry

3.3.3 UC_BrowseSearch

Use Case ID:	UC_03
Primary Actor	Visitor
Use Case Name:	UC_BrowseSearch
Description:	In this use case, the visitor uses the search functionality to find hotels based on their preferences such as location, price range, and facilities. The system displays a list of relevant hotels based on the search criteria.
Preconditions:	The visitor must be on the website and have access to the search functionality.
Basic Flow:	• Visitor navigates to the search bar. • Visitor enters a specific location in the search bar. • Visitor submits the search query by clicking on “Lets Explore” button. • System retrieves and displays a list of hotels in the specified location with hotel names, prices, and score ratings. • Each hotel listing includes a button to view detailed information about the hotel. • Visitor selects options for price range and facilities. • System filters and updates the list of hotels based on the selected price range and facilities.
Exceptional Flows:	• If no hotels match the search criteria, the system displays a message indicating that “No Recommended Hotel”
Postconditions:	The visitor can view a list of hotels matching their search criteria and filter options.

3.3.4 UC_ViewHotelDetails

Use Case ID:	UC_04
Primary Actor	Visitor
Use Case Name:	UC_ViewHotelDetails
Description:	This use case allows a visitor to view detailed information about a specific hotel after searching and selecting it.
Preconditions:	The visitor must have performed a search for hotels and viewed a list of search results.
Basic Flow:	• Visitor views a list of searched hotels. • Visitor clicks on the "Detail" button for a specific hotel. • System navigates the visitor to the hotel's detail page. • System fetches the hotel's detailed information from the database. • Hotel details, including room types, prices, and other relevant information, are displayed on the detail page.
Exceptional Flows:	• If the hotel details cannot be fetched (e.g., due to a database error), an error message is displayed.
Postconditions:	The visitor can view comprehensive details about the selected hotel, including room types, prices, and any other relevant information.

3.3.5 UC_LiveLocationSearch

Use Case ID:	UC_05
Primary Actor	Visitor
Use Case Name:	UC_LiveLocationSearch
Description:	In this use case, the visitor uses the live location feature on the website to find nearby hotels based on their current location. The system will display nearby hotels, and allow filtering by price and facilities.
Preconditions:	The visitor must be using a device with location services enabled and must grant permission for location access.
Basic Flow:	• User navigates to the live location search feature on the website. • System prompts the user to allow access to their current location. • User grants permission for location access. • System detects the user's current location. • System retrieves and displays hotels near the user's current location. • User can filter the list of nearby hotels based on price facilities.
Exceptional Flows:	• User denies permission to access their location. • Location services are not enabled on the device.
Postconditions:	After accessing their location, the user is shown a list of nearby hotels and can apply filters to refine their search. If permission is denied or location services are not enabled, the user receives appropriate messages and instructions.

3.3.6 UC_ListHotel

Use Case ID:	UC_06
Primary Actor	Visitor
Use Case Name:	UC_ListHotel
Description:	In this use case, a visitor lists a new hotel by providing the necessary details through a form on the website. The system then sends a notification email to the admin for review of the new listing.
Preconditions:	The visitor must be logged in and have access to the hotel listing form.
Basic Flow:	• Visitor navigates to the "List Property" option on the website. • Visitor accesses the hotel listing form. • Visitor fills in the required hotel details (e.g., hotel name, location, description, facilities, etc.). • Visitor submits the listing form. • System processes the form submission and stores the details. • System sends a notification email to the admin indicating a new hotel listing request is pending for review.
Exceptional Flows:	• The visitor submits incomplete or invalid hotel details. • The system displays an error message and prompts the visitor to correct the errors.
Postconditions:	After submission, the hotel listing request is recorded and a notification email is sent to the admin for review. The visitor receives a confirmation that their listing has been submitted successfully.

3.3.7 UC_ReviewSubmission

Use Case ID:	UC_07
Primary Actor	Visitor
Use Case Name:	UC_ReviewSubmission
Description:	In this use case, a visitor submits a review for a specific hotel. The system first checks if the user is registered and logged in. If not, the user is redirected to the login page. After successful login, the user can submit their review.
Preconditions:	The user must be registered and logged in to submit a review.
Basic Flow:	• User navigates to the detail page of a specific hotel. • User locates the review submission section. • System checks if the user is logged in. • If the user is not logged in, they are redirected to the login page. • User logs in using valid credentials. • Upon successful login, the user is redirected back to the hotel detail page. • User enters their review for hotel. • User submits the review by clicking the submit button. • System processes and saves the review in the database. • A confirmation message is displayed to the user.
Exceptional Flows:	• If the user is not registered, the system prompts for registration before login the system for review. • If login credentials are incorrect, the user is shown an error message and prompted to retry.
Postconditions:	The user's review is successfully submitted and saved in the system. The user is notified of the successful submission. If login was required, the user is granted access to review submission only after logging in.

Use_Case_Diagram of HotelHype

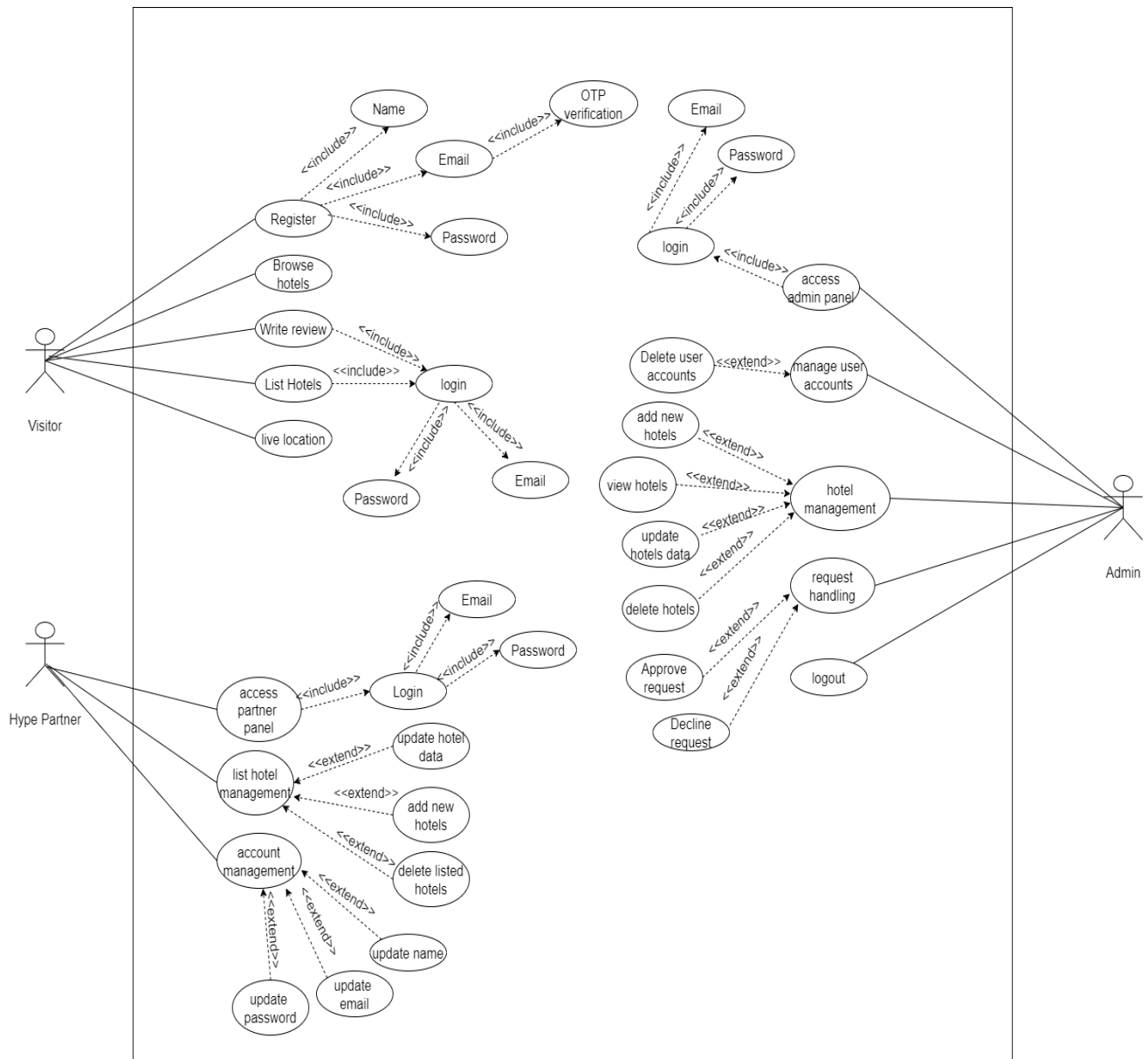


Figure 4: UCD of HotelHype

4 SPECIFIC REQUIREMENTS

4.2 Website Features

4.2.1 Registration

The user must provide information such as a username, phone number, and email address and password. All the information will be stored on MySQL database. This information will be used by the website for verifying the user and validating logins.

4.2.2 Login Account

In order to use website, visitor does not need login for searching but at some points, visitor, Partner and admin must logged in to his account.

4.2.3 Hype Partner Account

The users start conversation with others by searching user from the user's fragment using their username or email id. Users set timer before sending message. Timer starts only after receiver has seen the message and the messages disappear after the timer's expiration.

4.2.4 Encryption

All the Passwords during registration are encrypted. The passwords are encrypted using an encryption code by python before storing in database.

4.2.5 Live Location Feature

The users can also avail the feature of searching hotels with live location feature to quick the searching recommended hotels.

4.2.6 Search Recommended Hotels

Along with searching users with their preferences (city, facilities and prices), the users can get the recommended hotels according to their searching.

4.2.7 Write Review

The user can also write reviews on hotel to express their opinions. By logging in to their accounts.

4.2.8 Partner Panel

The visitor can also be a partner of our website. By just filling in the hotel hype listing form, after the account activation, that person be our hotel hype partner and by logging in to the account, partner can access to their dashboard panel.

4.2.9 Admin Panel

Our website provide the admin panel for the administrators to control all the website functionalities and partner accounts to smoothly and efficiently. Through admin panel, admin can handle all the functionalities with just one click.

5 QUALITY ATTRIBUTES

5.2 Usability

Usability requirements deal in this Website. It is designed for easy access from any device with active internet connection. It helps to create better and useful user interface then it maintains functionality and usability.

5.3 Performance

Its quality performance should be successful and error free. It come with complete functional and non-functional requirements. This web application must work properly according to the requirements. It should consume less time and cost. That's why it has completed performance quality attribute.

5.4 User-Friendly Interface

Our Web application has user-Friendly interface. Users can easily interact with the application easily.

5.5 Security

In our website, passwords are encrypted during registration and emails are send for verification to protect against unauthorized access. It also have the transient nature to make this website more secure.

5.6 Readability

Our web application considers the readability requirements for user. The content in our website has a readability feature and easy to understand. Every section of website is designed in a user-friendly way to make it more readable and understandable.

5.7 Browser Version

Our application will look stunning on any update version of browser. This feature will enable our website to work smoothly on any kind of devices with active internet connection.

6 EXTERNAL INTERFACE REQUIREMENTS

6.2 User Interfaces

The interface is in the form of Web application. It is designed to be functional and minimal in its styling. Html, CSS and JS is used to design layouts and user interactivity of our website, which will make the user interface user friendly. All the core functionality of an website are built using Python.

6.3 Software Interfaces

Visual studio code is used as software interface to design website application user-friendly . MySQL is used as a database to store data of Web Application. MySQL is relational database that sync user's data in real time.

6.4 Hardware Interfaces

As it is a web application, therefore any device with an active internet connection would be necessary to run this web application. Below are the minimum requirements of website.

1. Browser Compatibility (Google Chrome, Mozilla Firefox, Safari, Microsoft Edge).
2. 4 GB RAM for smooth performance.
3. Dual-core processor.
4. 1366x768 pixels or higher for optimal user experience.

7 DESIGN METHODOLOGY

We are using ‘Agile’ methodology also known as the Agile Model is a iterative software development process, Agile methods break tasks into smaller iterations, or parts do not directly involve long term planning. This involves fully documenting a project in advance, including the user interface, user requirements, and all the features’ variations and outcomes.

We are using Agile method techniques in our projects.

- Requirement analysis
- Design the requirements
- Construction / Iteration
- Testing / Quality Assurance
- Deployment
- Feedback

7.1 Agile Model Diagram

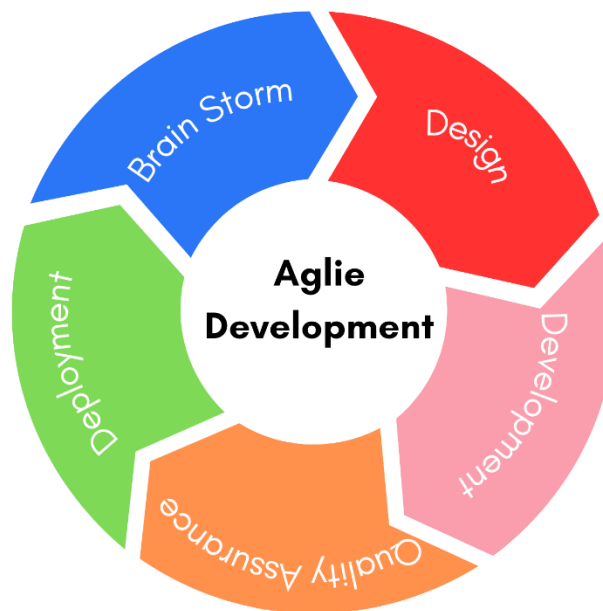


Figure 5: Agile Model

8 PROJECT GANTT CHART

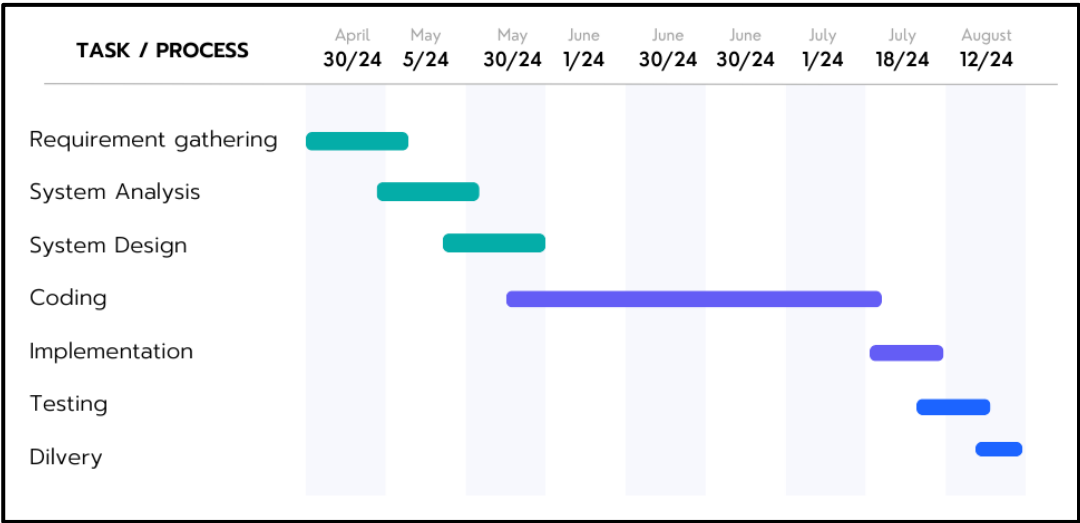


Figure 6: Gantt Chart

9 DESIGN MODELS

The data model is a subset of the implementation model, which describes the logical and physical representation of persistent data in the system. (R3, n.d.)

9.2 Class Diagram

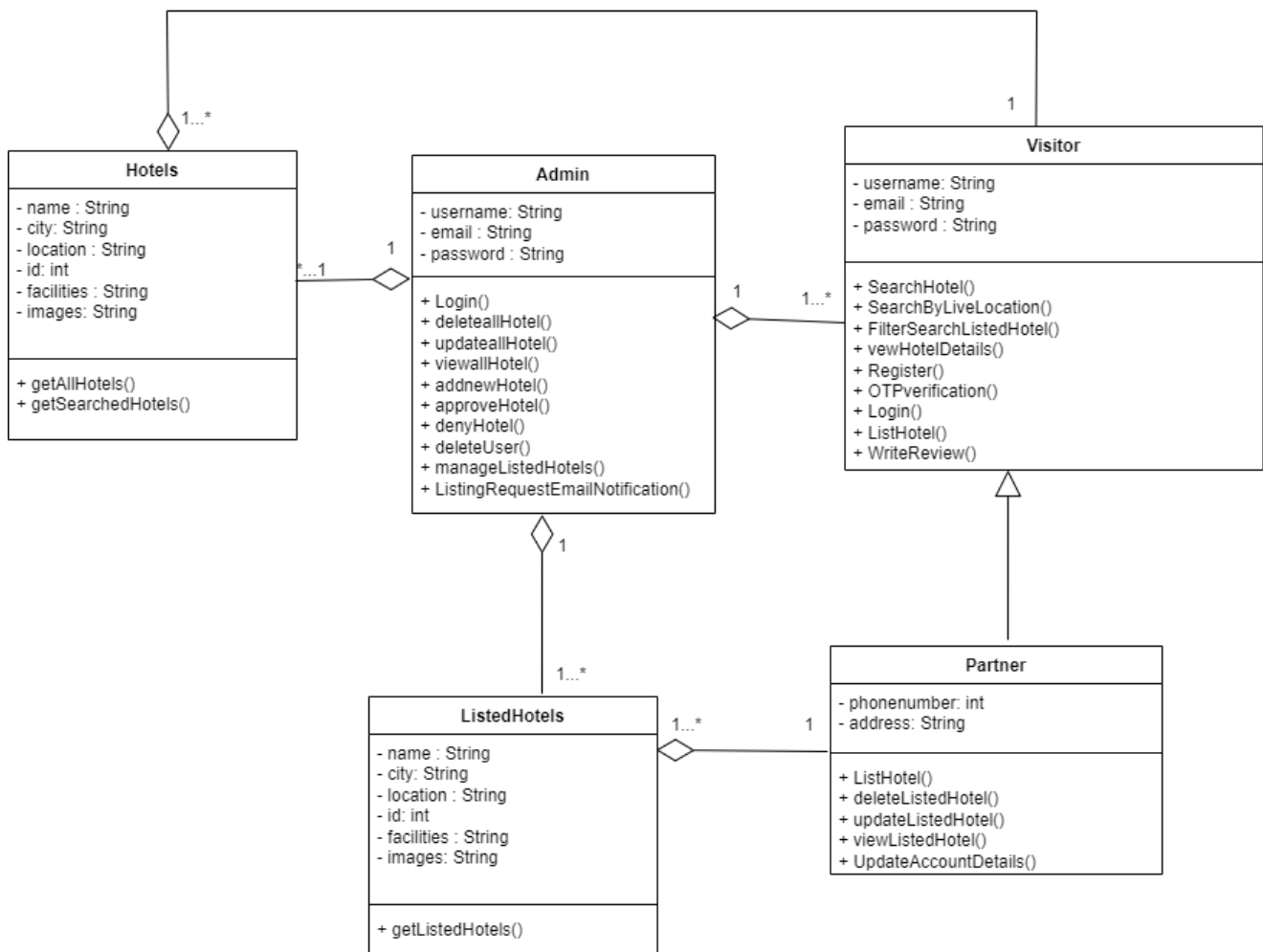


Figure 7: Class Diagram of HotelHype

9.3 Collaboration Diagram

9.2.1 Admin Collaboration Diagram

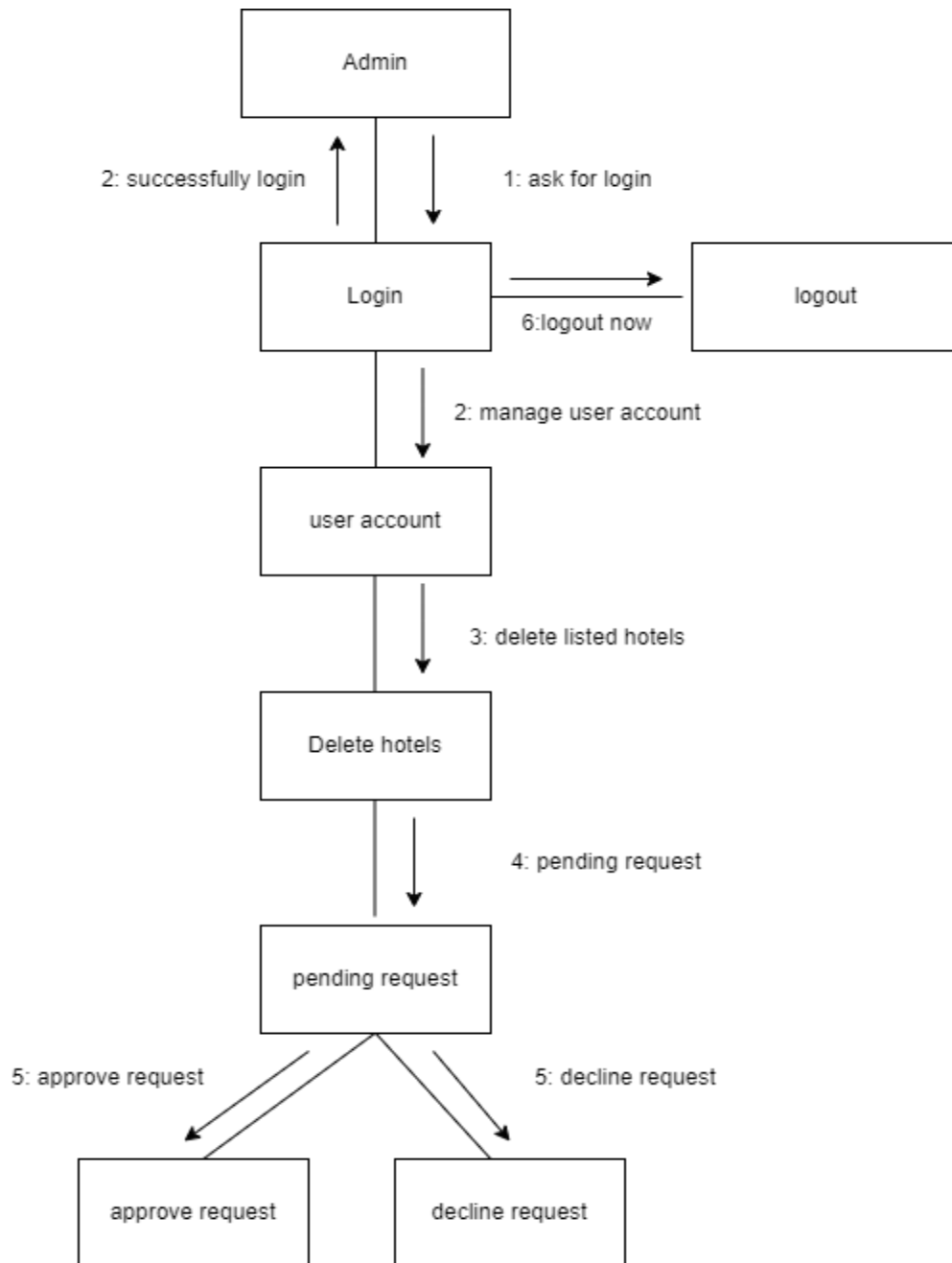


Figure 8: Collaboration Diagram of Admin

9.2.2 Partner Collaboration Diagram

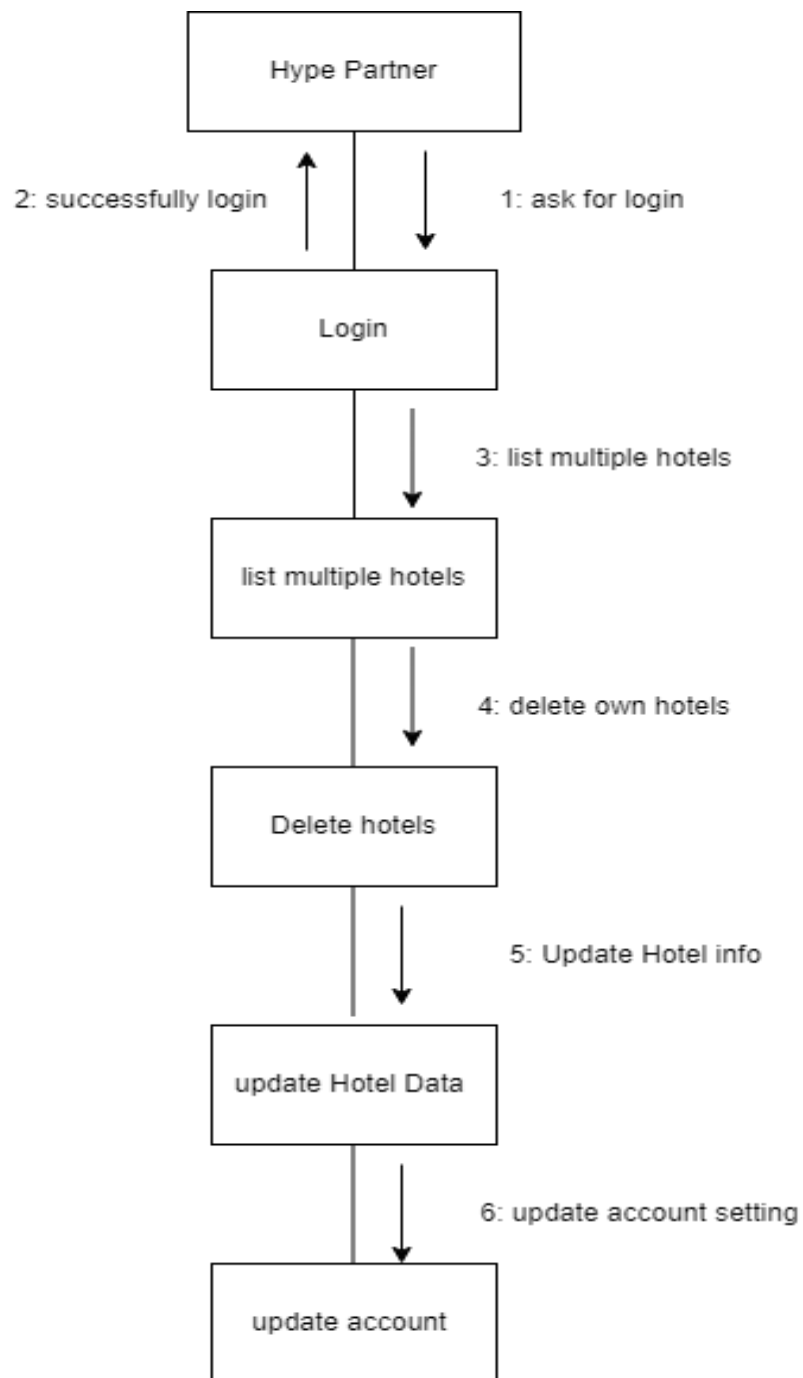


Figure 9: Collaboration Diagram of Partner

9.2.3 Visitor Collaboration Diagram

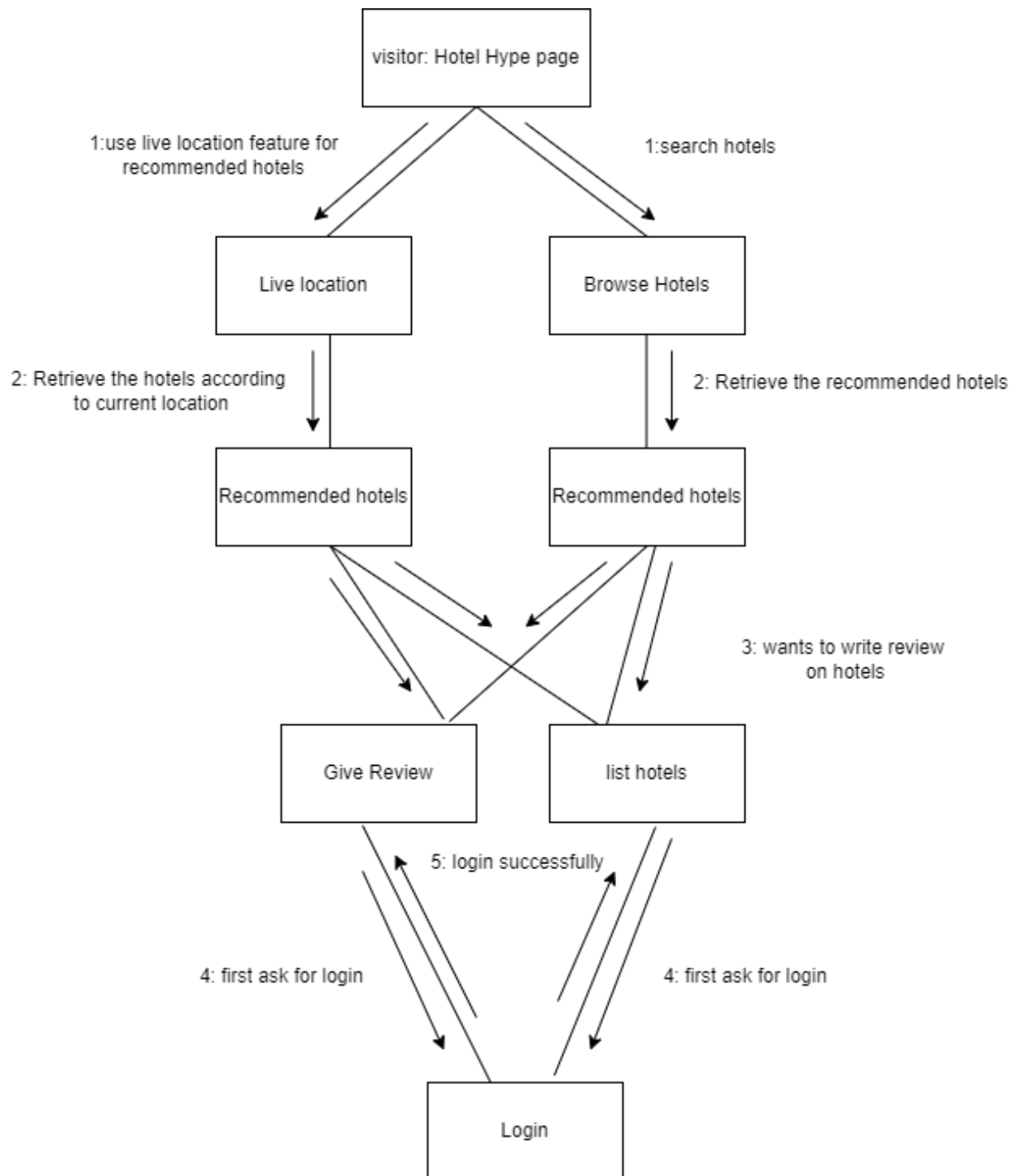


Figure 10: Collaboration Diagram of Visitor

9.3 Sequence Diagram

9.3.1 Admin Login and User Management Sequence Diagram

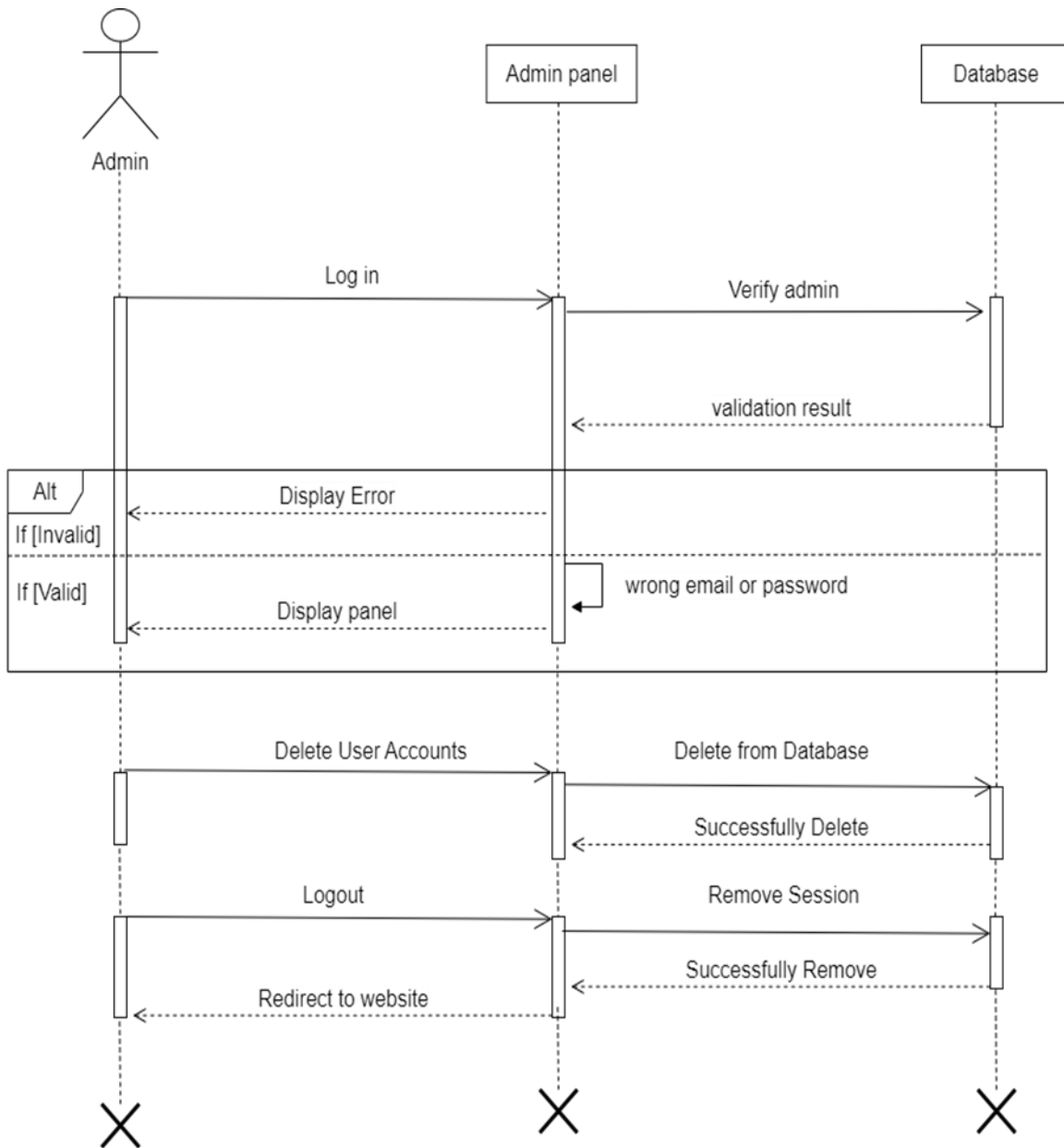


Figure 11: Sequence Diagram of Admin Login

9.3.2 Admin Hotel Management Sequence Diagram

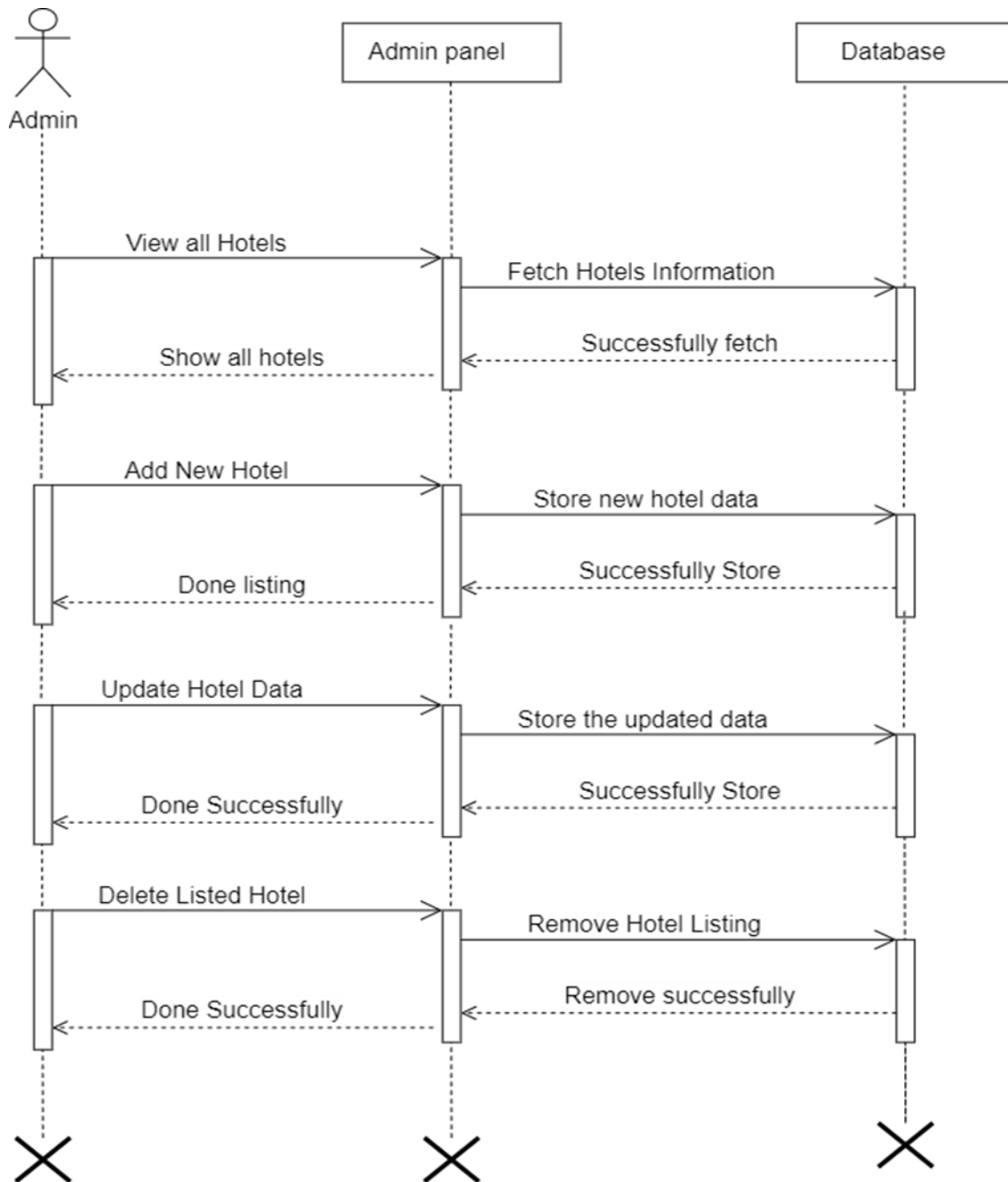


Figure 12: Sequence Diagram of Admin Hotel Management

9.3.3 Partner Login Sequence Diagram

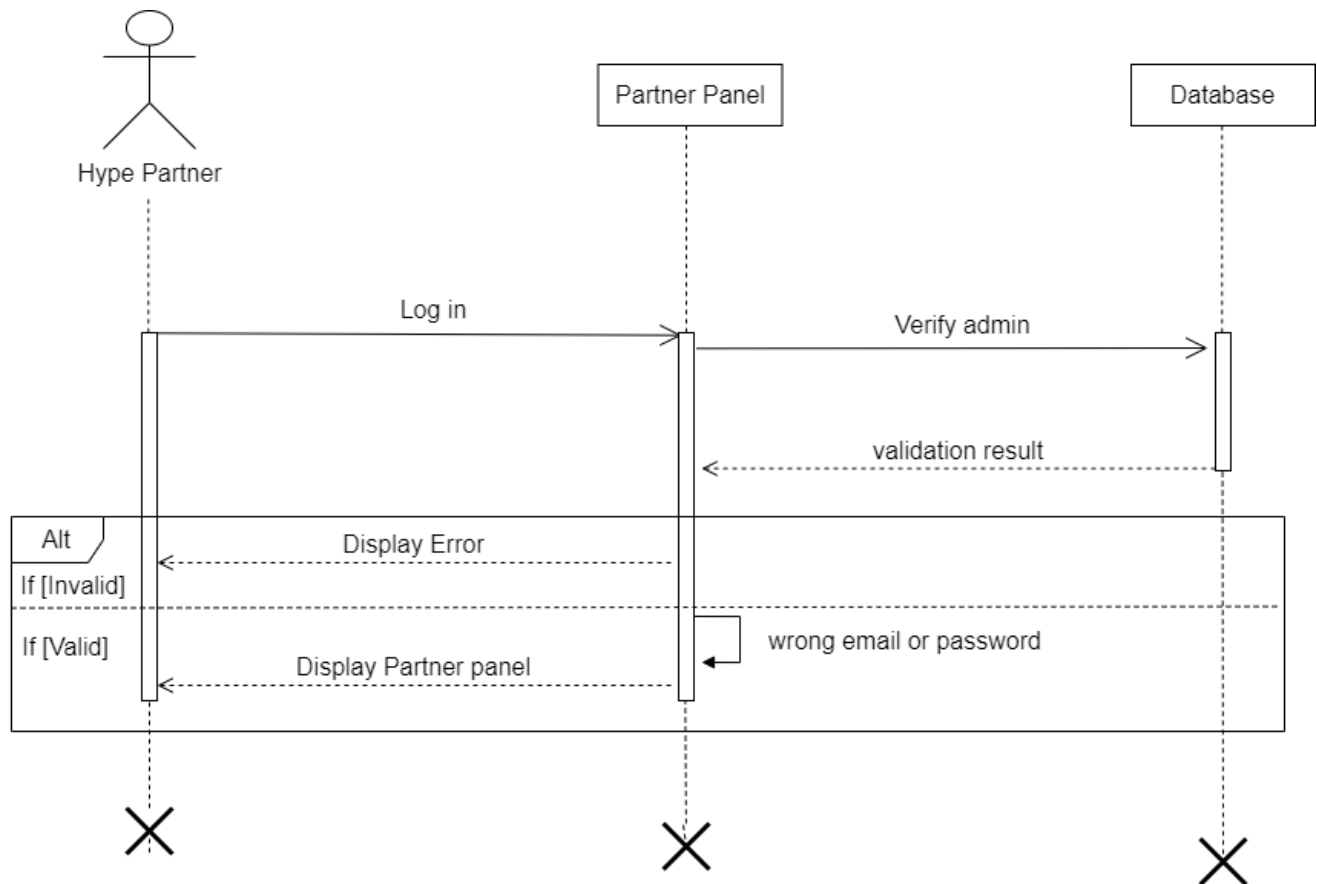


Figure 13: Sequence Diagram of Partner Login

9.3.4 Partner Listed Hotel Management Sequence Diagram

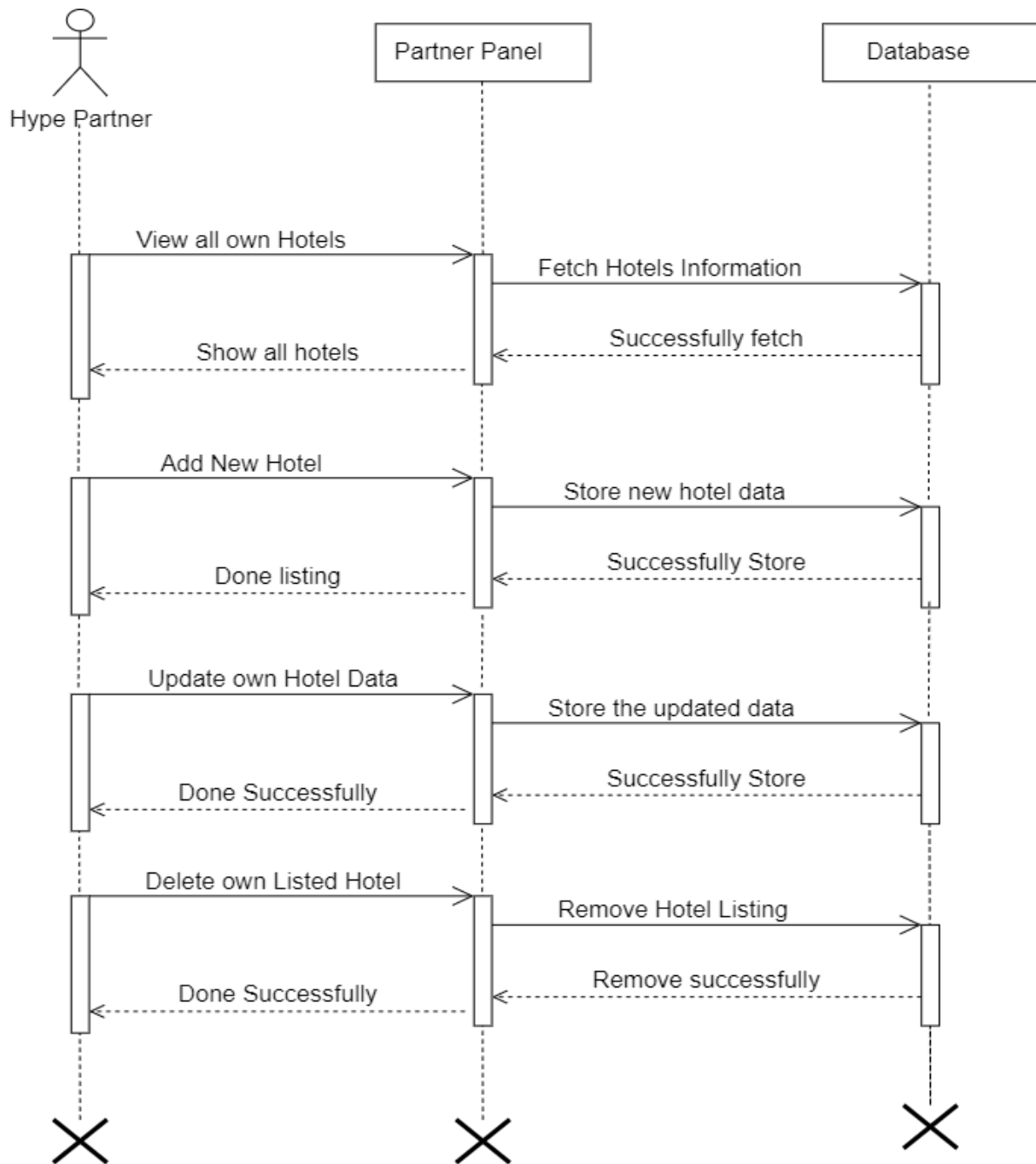


Figure 14: Sequence Diagram of Partner Listed Hotel Management

9.3.5 Partner Account Management Sequence Diagram

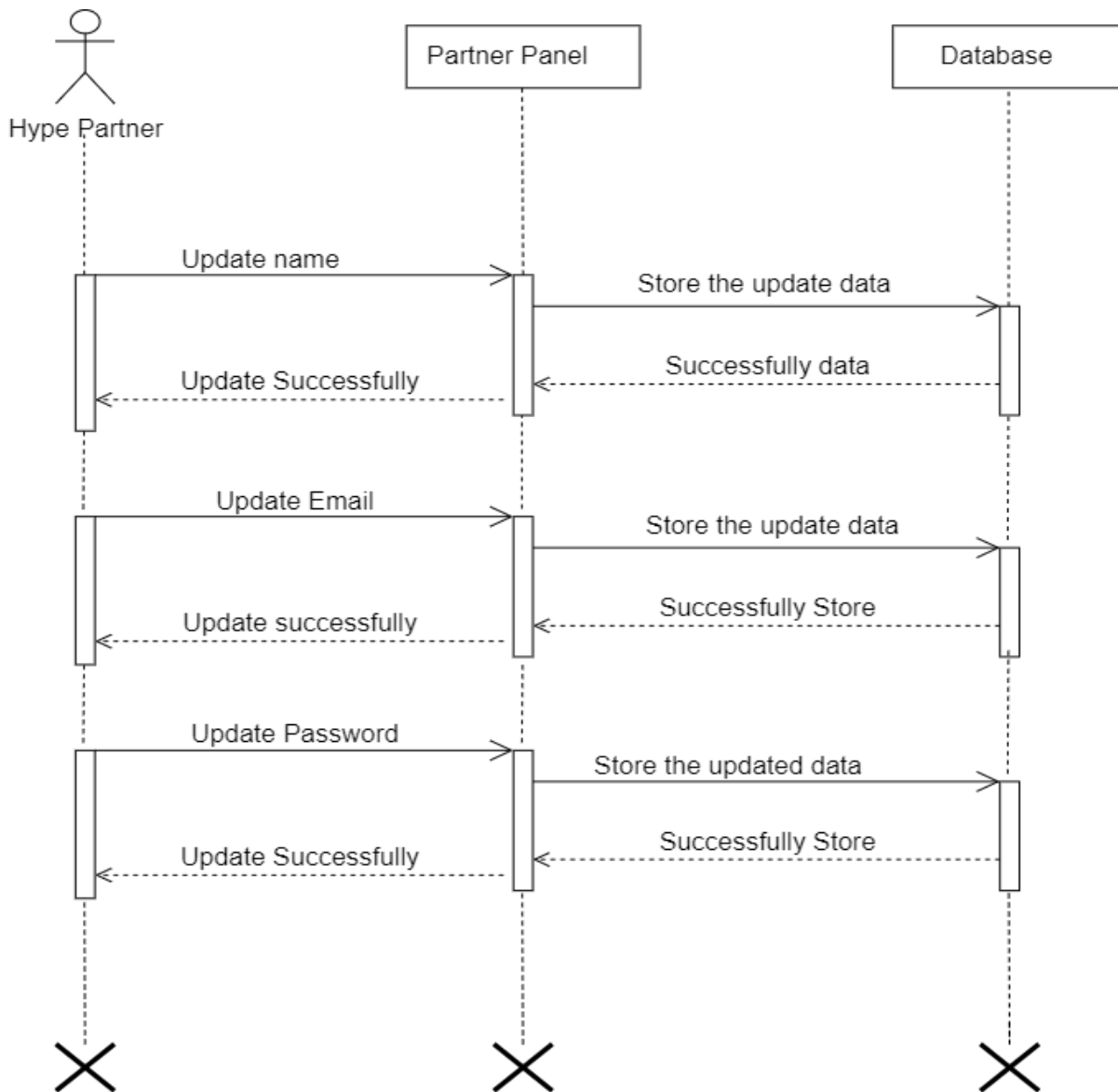


Figure 15: Sequence Diagram of Partner Account Management

9.3.6 Visitor Login Sequence Diagram

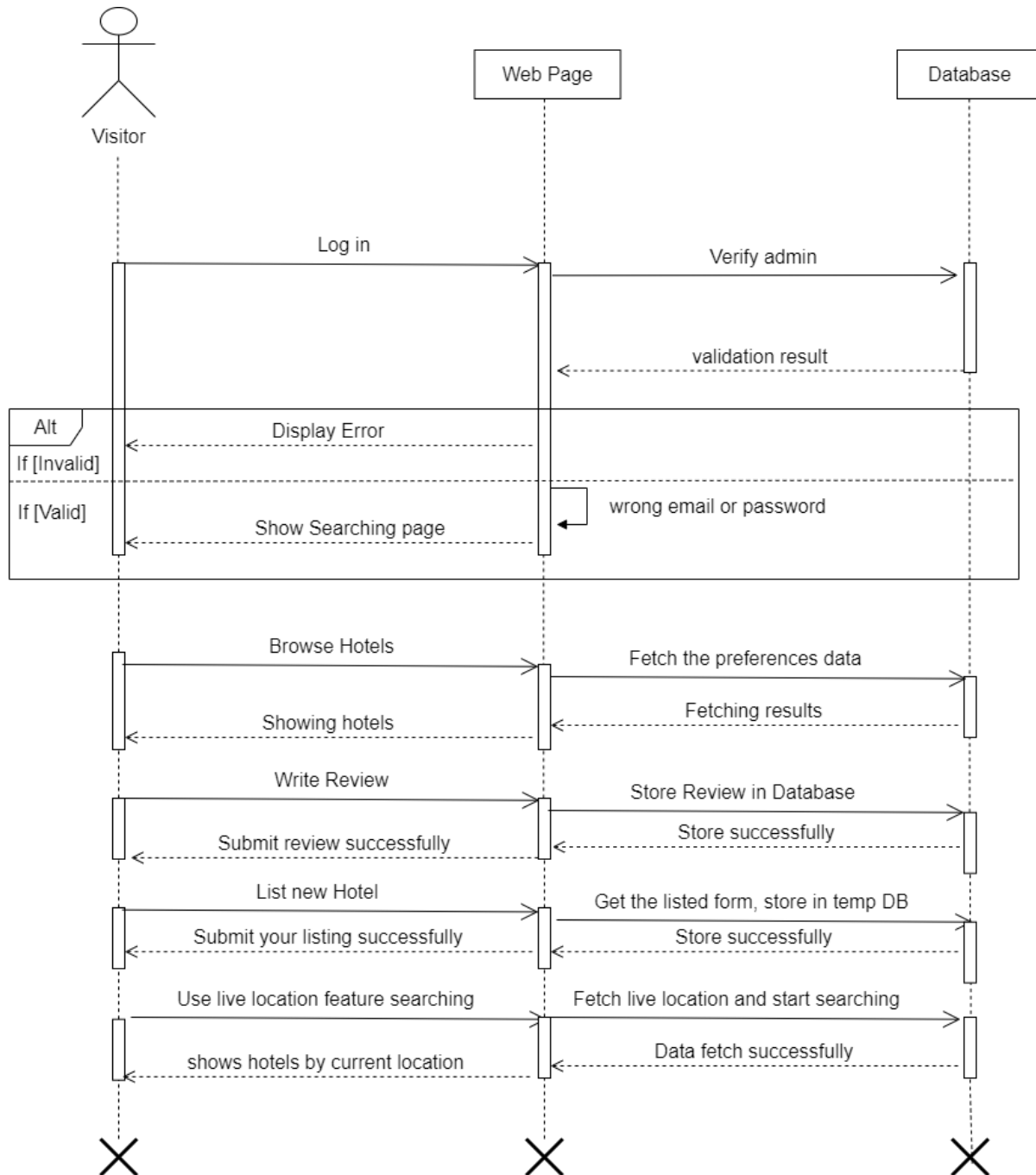


Figure 16: Sequence Diagram of Visitor

9.4 Activity Diagrams

9.4.1 Admin Activity Diagram

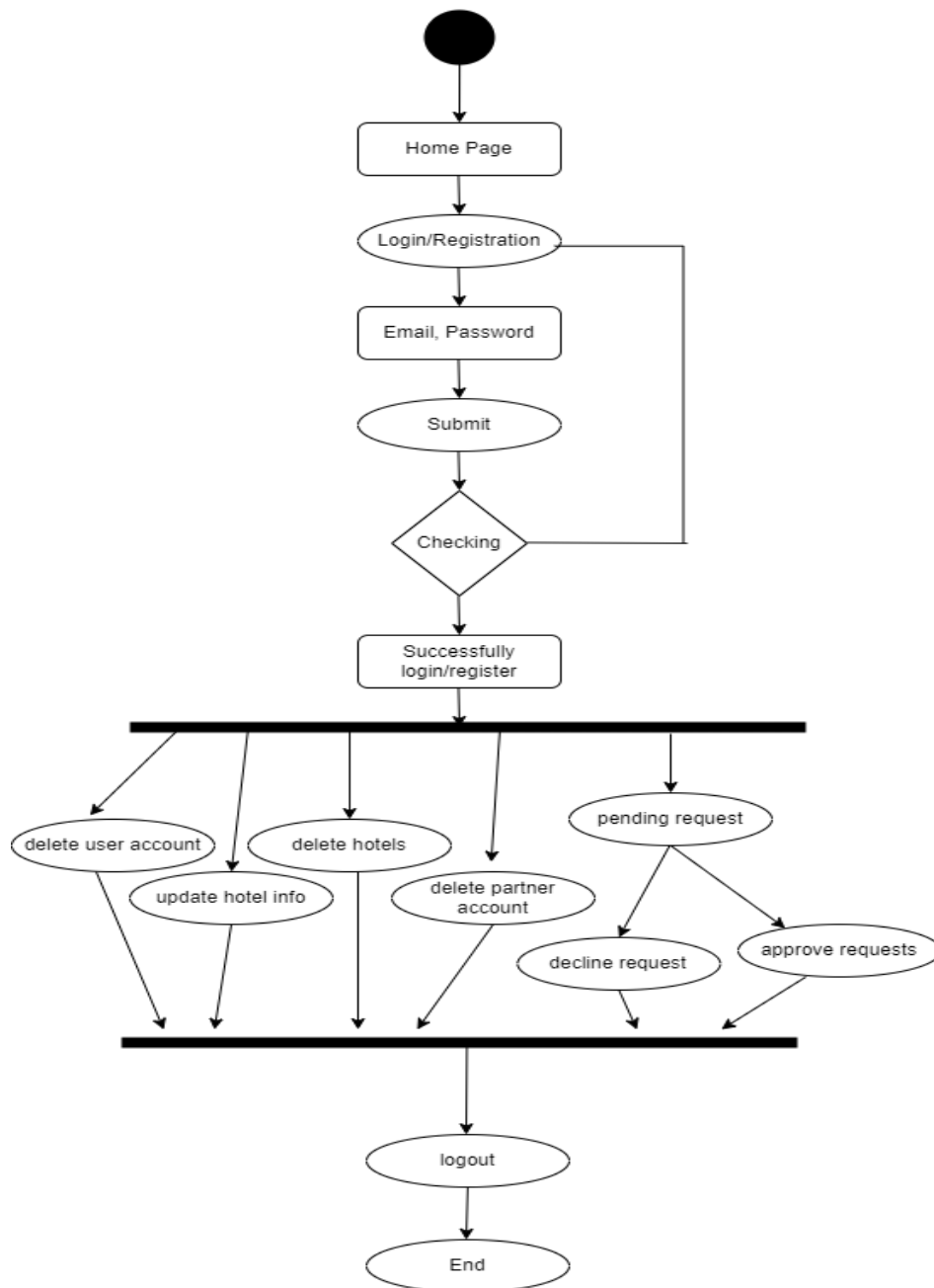


Figure 17: Activity Diagram of Admin

9.4.2 Partner Activity Diagram

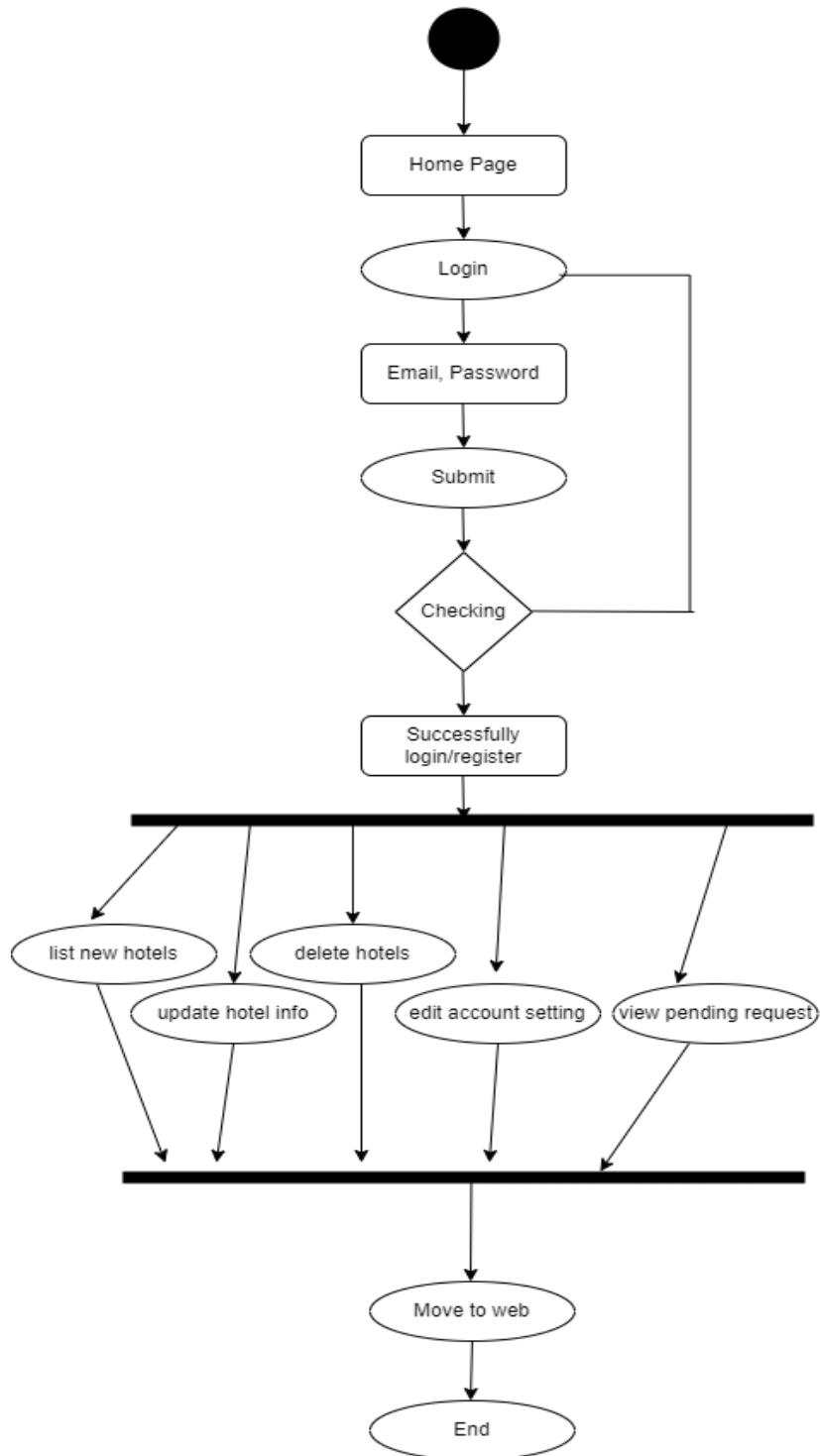


Figure 18: Activity Diagram of Partner

9.4.3 Visitor Activity Diagram

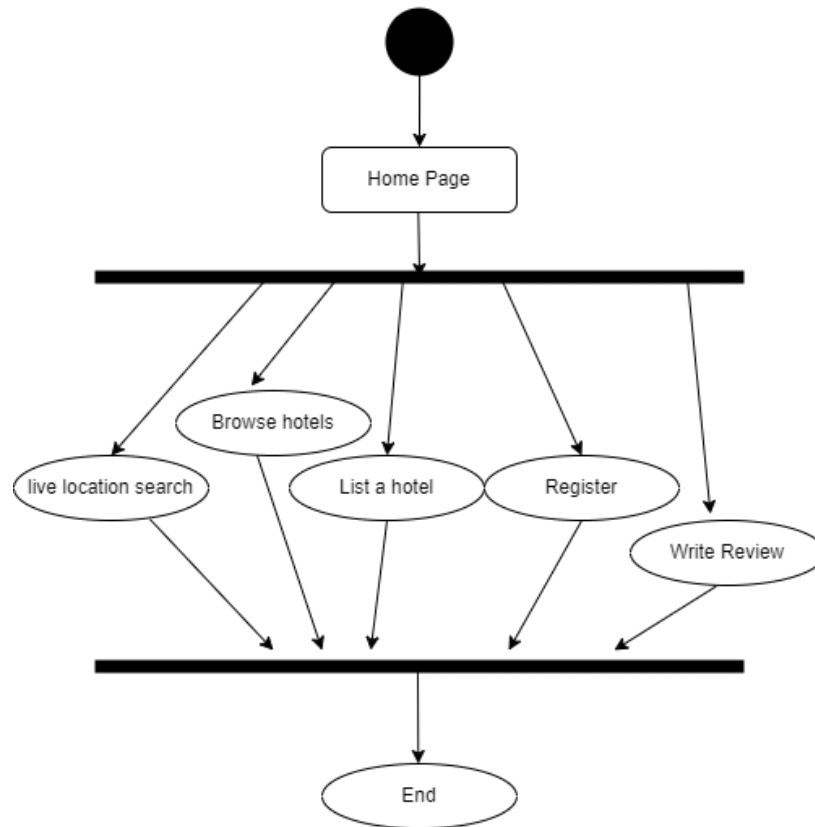


Figure 19: Activity Diagram of Visitor

9.5 DFD Diagrams

9.5.1 DFD (Level 0)

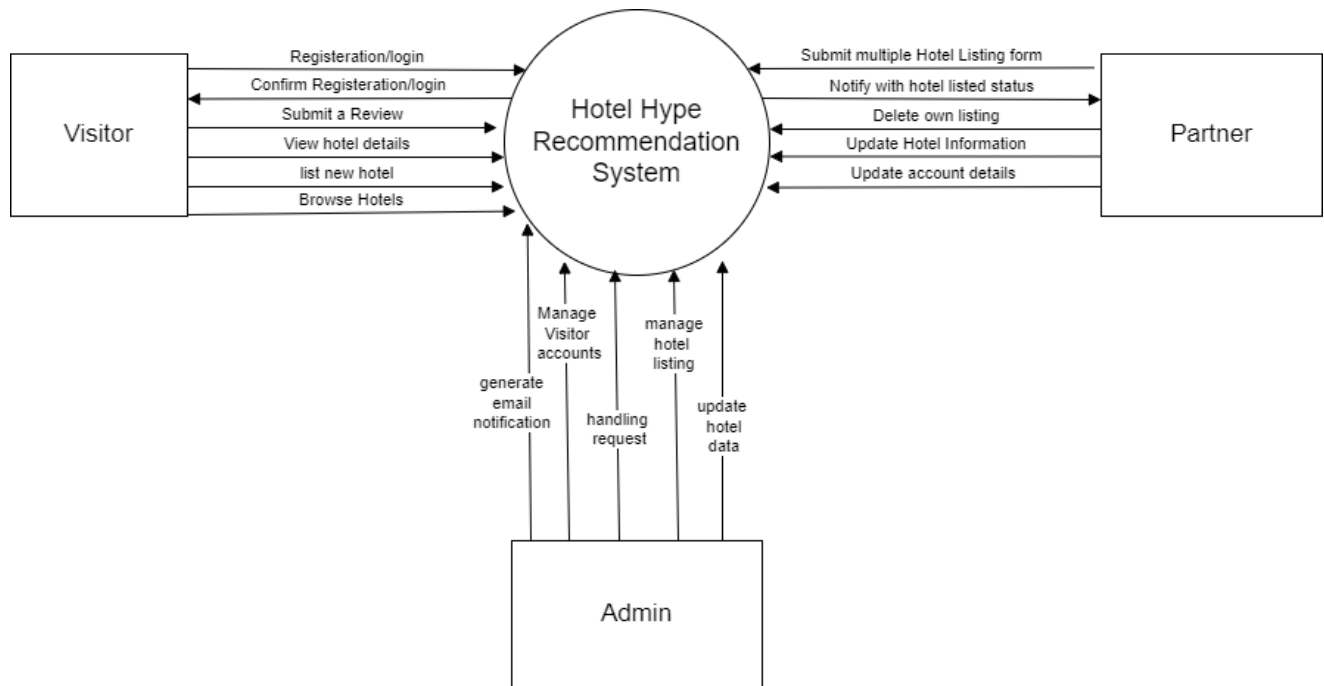


Figure 20: DFD Level 0

9.5.2 DFD (Level 1)

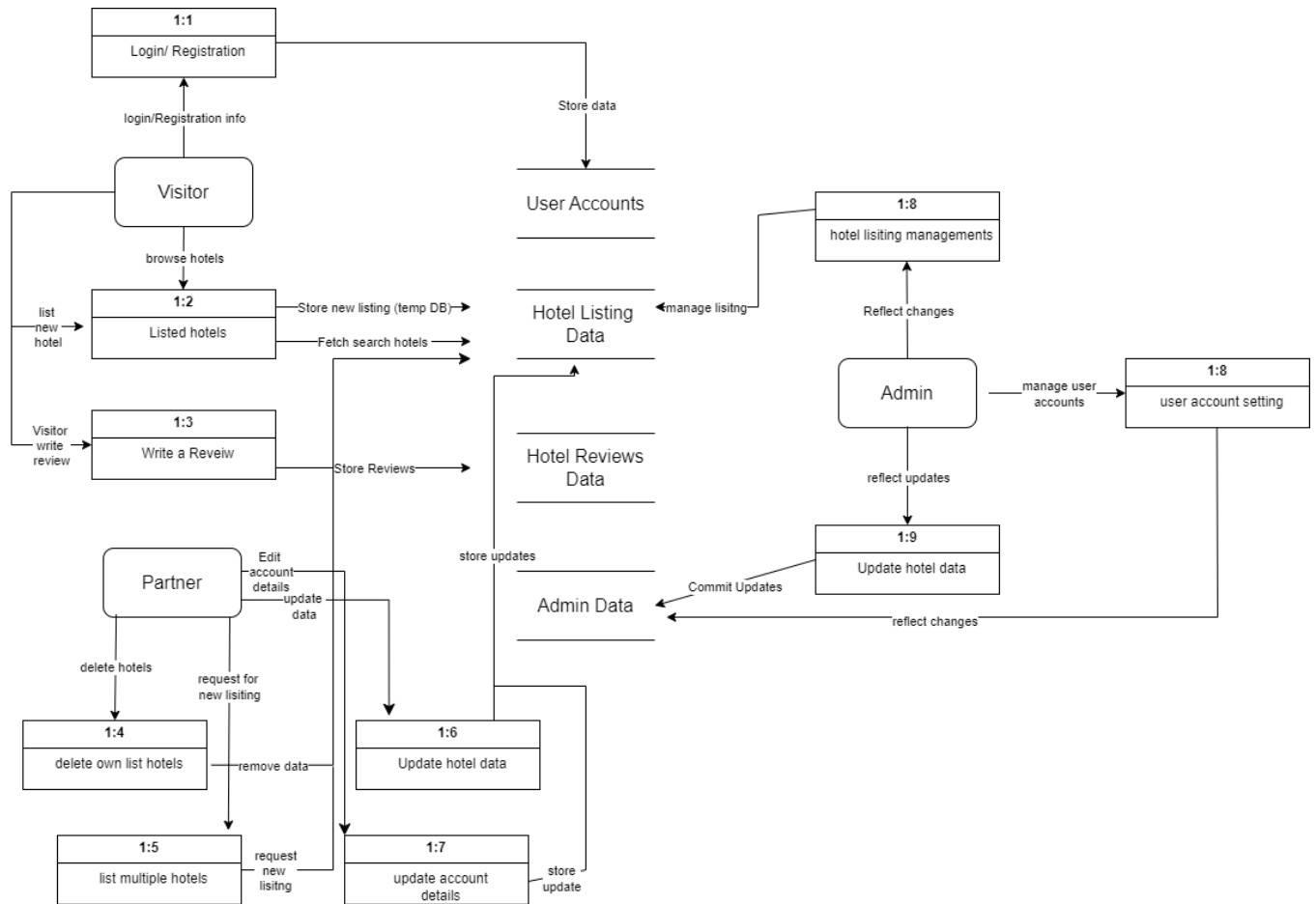


Figure 21: DFD Level 1

10 APPLICATION ARCHITECTURE

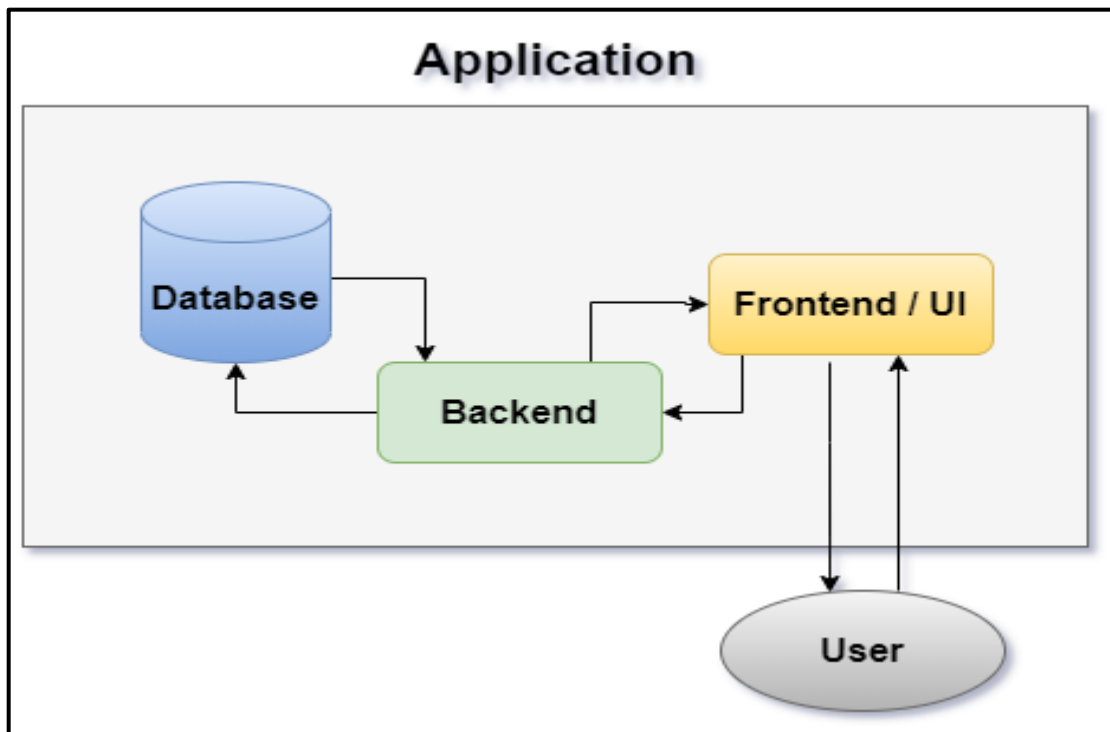


Figure 22: Application Architecture

11 USER INTERFACE

The “Hotel Hype” is Web application running over all devices. The application is developed using Front-end and back-end technologies. The interface is in the form web application. Html, CSS, JS is used to setup the web design layout and add styling to make the interface user friendly. To develop backend of this application we use Python with Flask framework, MYSQL is used as a database that is relational database. We create hotel search page, login and registration activity for user, hotel owners profile page and admin page. A hotel listing activity to send request to admin for listing hotels. We do sentiment analysis for hotel reviews for finding the rating score to recommended hotels to users. User writing review activity takes the reviews from users and pass through our sentiment analysis and recommendation model process to update the current rating score of that hotel. A panel activity to display profile of currently logged hotel owner.

12 TESTING

12.1 Introduction to Test Cases

Testing is a critical phase in the software development lifecycle that aims to ensure the quality and functionality of software applications. It involves executing the software to identify defects, verify that it meets specified requirements, and ensure it behaves as expected under various conditions.

12.1.1 Objective of Testing

- Detect and document defects or bugs in the software to be fixed before release.
- Ensure that the software meets the specified requirements and performs the intended functions correctly.
- Enhance the quality and reliability of the software by finding and addressing issues early

12.2 Admin Test Cases

12.2.1 Admin Login

Test Case ID	Case type	Expected Result	Actual Result	Pass / Fail
TC01	Wrong email and password	The application throws an error and prevent login.	Application displays "Invalid email or password."	Pass
TC02	Enter correct email and password	The application logs in successfully and redirects to the admin dashboard.	Application redirects to the admin dashboard with a message "Logged in Successfully."	Pass

12.2.2 Hotel Management

12.2.2.1 View Hotels

Test Case ID	Case type	Expected Result	Actual Result	Pass / Fail
TC03	Admin can views Hotels	Application displays a list of existing hotels.	List of hotels is displayed with their score rating and location.	Pass

12.2.2.2 Add New Hotels

Test Case ID	Case type	Expected Result	Actual Result	Pass / Fail
TC04	Add new hotels with valid and all required adta	New hotel is added successfully,confirmation message is displayed on screen	Confirmation message is displayed and hotel is added.	Pass
TC05	Add New Hotel with Missing Data	The application displays error message indicating missing information.	Error message displayed for missing fields.	Pass

12.2.2.3 Delete Hotels

Test Case ID	Case type	Expected Result	Actual Result	Pass / Fail
TC06	Delete Hotel	The hotel is deleted successfully, and a confirmation message is displayed.	Confirmation message is displayed, and hotel is removed from the database too.	Pass

12.2.3 User Management

12.2.3.1 Delete User or Listed partner accounts

Test Case ID	Case type	Expected Result	Actual Result	Pass / Fail
TC07	Delete Listed Hotel Partner Account also registered user accounts too	The account is deleted successfully,confirmation message is displayed.	Confirmation message is displayed, and account is removed.	Pass

12.2.4 Request Handling

12.2.4.1 Email Notification for Listing Request

Test Case ID	Case type	Expected Result	Actual Result	Pass / Fail
TC08	Email Notification on Listing Request	Admin receives an email notification from system about a new listing request.	Admin receives an email with details of the listing request.	Pass

12.2.4.2 Approve/Deny Listing request

Test Case ID	Case type	Expected Result	Actual Result	Pass / Fail
TC08	Approve Hotels Partner Listing Request	The hotel listing request is approved successfully, and confirmation message is displayed.	Confirmation message is displayed, and listed data is added into the database with partner details stored	Pass
TC09	Deny Listing Request	The hotel listing request is denied successfully, and confirmation message is displayed.	Confirmation message is displayed, and hotel is denied from storing in database	Pass

12.3 Hotel Partner Test Cases

12.3.1 Browse Through Website

Test Case ID	Case type	Expected Result	Actual Result	Pass / Fail
TC01	Search by Location	Application displays a list of hotels based on the searched location.	List of hotels matching the location is displayed.	Pass
TC02	Search by Price	The application displays a list of hotels within the specified price range.	List of hotels within the price range is displayed.	Pass
TC03	Search by Facilities	The application displays a list of hotels based on the selected facilities.	List of hotels with selected facilities is displayed.	Pass
TC04	Search by Live Location	The application displays hotels near the user's current location.	List of hotels near the live location is displayed.	Pass

12.3.2 Login and Access to Partner Panel

12.3.2.1 Login with Email

Test Case ID	Case type	Expected Result	Actual Result	Pass / Fail
TC05	Login with Valid Credentials	The application logs in successfully and redirects to the partner panel.	Partner panel is displayed after login	Pass
TC06	Login with Invalid Credentials	The application displays an error message.	Error message indicating “with this email no partner is Listed”	Pass

12.3.2.2 Access Partner Panel

Test Case ID	Case type	Expected Result	Actual Result	Pass / Fail
TC06	View Listed Hotels	The partner panel displays count of listed hotels.	Count of listed hotels is displayed.	Pass
TC07	View Pending Requests	Partner panel displays count of pending requests.	Count of pending hotels is displayed.	Pass
TC08	View Approved Requests	Partner panel count of approved requests.	Count of approved hotels is displayed.	Pass

TC09	View Denied Requests	The partner panel displays all denied requests.	List of denied requests is displayed.	Pass
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12.3.3 Hotel Management

12.3.3.1 List a new Hotel

Test Case ID	Case type	Expected Result	Actual Result	Pass / Fail
TC01	Submit Hotel Listing	The hotel listing request is sent to the admin for approval, with correctly filled data	Confirmation message is displayed “data is submitted for review” and request is sent.	Pass
TC02	Submit Incomplete Listing	Application displays an error message indicating missing information.	Error message is displayed for missing fields.	Pass

12.3.3.2 Update existing hotel

Test Case ID	Case type	Expected Result	Actual Result	Pass / Fail
TC03	Update Hotel Details	The hotel details are updated successfully, and a confirmation message is displayed.	Confirmation message is displayed, “Hotel details are updated.”	Pass

TC04	Update using an Invalid Data	The application displays an error message indicating invalid field data.	Error message is displayed for invalid data.	Pass
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12.3.3.3 Register and List Hotels with Multiple Emails

Test Case ID	Case type	Expected Result	Actual Result	Pass / Fail
TC05	Register with New Email	The application allows registration with a new email.	Registration is successful with new email.	Pass
TC06	List Hotels with New Email	The partner can list hotels using the newly registered email.	Hotels are listed successfully with new email and Partner get access to new panel registered with that email	Pass

12.3.3.4 Email Notification on Approval/Deny

Test Case ID	Case type	Expected Result	Actual Result	Pass / Fail
TC05	Email Notification on Approval	The user receives an email confirming that their hotel listing has been approved.	User receives an email stating the hotel listing was approved.	Pass

TC06	Email Notification on Denial	The user receives an email informing them that their hotel listing has been denied.	User receives an email stating the hotel listing was denied.	Pass
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12.3.4 Account Management

Test Case ID	Case type	Expected Result	Actual Result	Pass / Fail
TC05	Change Email	The email is updated successfully, and a confirmation message is displayed.	Confirmation message is displayed, “email is updated successfully”	Pass
TC06	Change Username	The username is updated successfully, and a confirmation message is displayed.	Confirmation message is displayed, “username is updated successfully”	Pass
TC06	Change Password	The password is updated successfully, and a confirmation message is displayed.	Confirmation message is displayed, “password is updated successfully”	Pass

12.4 Visitor Test Cases

12.4.1 Browse Through Website

Test Case ID	Case type	Expected Result	Actual Result	Pass / Fail
TC01	Search by Location	Application displays a list of hotels based on the searched location.	List of hotels matching the location is displayed.	Pass
TC02	Search by Price	The application displays a list of hotels within the specified price range.	List of hotels within the price range is displayed.	Pass
TC03	Search by Facilities	The application displays a list of hotels based on the selected facilities.	List of hotels with selected facilities is displayed.	Pass
TC04	Search by Live Location	The application displays hotels near the user's current location.	List of hotels near the live location is displayed.	Pass

12.4.2 Registration as a new User

Test Case ID	Case type	Expected Result	Actual Result	Pass / Fail
TC05	Register with Valid Details	The user is successfully registered, and a confirmation message is displayed.	Confirmation message is displayed, “ registration is successful.”	Pass
TC06	Register with Existing email	The application displays an error message indicating that the email is already in use.	Error message for duplicate email is displayed.	Pass
TC06	Register with Invalid details	The application displays an error message indicating missing or invalid fields.	Error message for invalid or missing details is displayed.	Pass

12.4.3 OTP Verification

Test Case ID	Case type	Expected Result	Actual Result	Pass / Fail
TC05	OTP Verification on Registration	The visitor receives an OTP via email, enters it correctly, and their email is verified.	Visitor receives OTP, enters it, and their email is verified. Message displayed that “email is verified”.	Pass
TC06	Invalid OTP Entry	The system displays an error message for incorrect OTP entries.	An error message is displayed when an incorrect OTP is entered.	Pass

12.4.4 Login to website

Test Case ID	Case type	Expected Result	Actual Result	Pass / Fail
TC05	Login with Valid credentials	The user is logged in successfully and redirected to the homepage	Successful login and redirection to the homepage.	Pass
TC06	Login with Unregistered Email	The application displays an error message indicating the email is not registered.	Error message displayed "Email is not registered"	Pass

12.4.5 Becoming hotel Partner

Test Case ID	Case type	Expected Result	Actual Result	Pass / Fail
TC05	List First Hotel	The user can list a hotel with valid details from website and a request is sent to the admin for approval. If the request is approved visitor now becomes hotel Partner	Confirmation message is displayed, "The request is sent."	Pass
TC06	List Hotel with Invalid Details	The application displays an error message indicating missing or invalid details.	Error message is displayed "Invalid or missing Details"	Pass

12.4.6 Accessing Partner Panel

Test Case ID	Case type	Expected Result	Actual Result	Pass / Fail
TC05	Access Partner Panel	After being approved as a listed partner, the user become the Partner and can access the partner panel.	Partner panel is accessible, and relevant information is displayed.	Pass

13 CRITICAL PATH METHOD (R5)

CPM models the activities and events of a project as a network. Activities are depicted as nodes on the network and events that signify the beginning or ending of activities are depicted as arcs or lines between the nodes. The following is an example of a CPM network diagram: Steps in CPM Project Planning

- Specify the individual activities.
- Determine the sequence of those activities.
- Draw a network diagram.
- Estimate the completion time for each activity.
- Identify the critical path (longest path through the network)
- Update the CPM diagram as the project progresses.

13.1 Specify the Individual Activities

Activity ID	Activity Description
A	Proposal
B	Project Scope
C	Project Planning and Analysis
D	High Level Use Case Description and Diagram
E	Designing and Modeling
F	Application Development
G	Database Development
H	MySQL Integration

I	Testing and Modification
J	Deployment

13.2 Determine the Sequence of the Activities

Activity ID	Dependency
A	None
B	A
C	A, B
D	C
E	D
F	E
G	E
H	E, F, G
I	F, G, H
J	I

13.3 Draw the Network Diagram (R6)

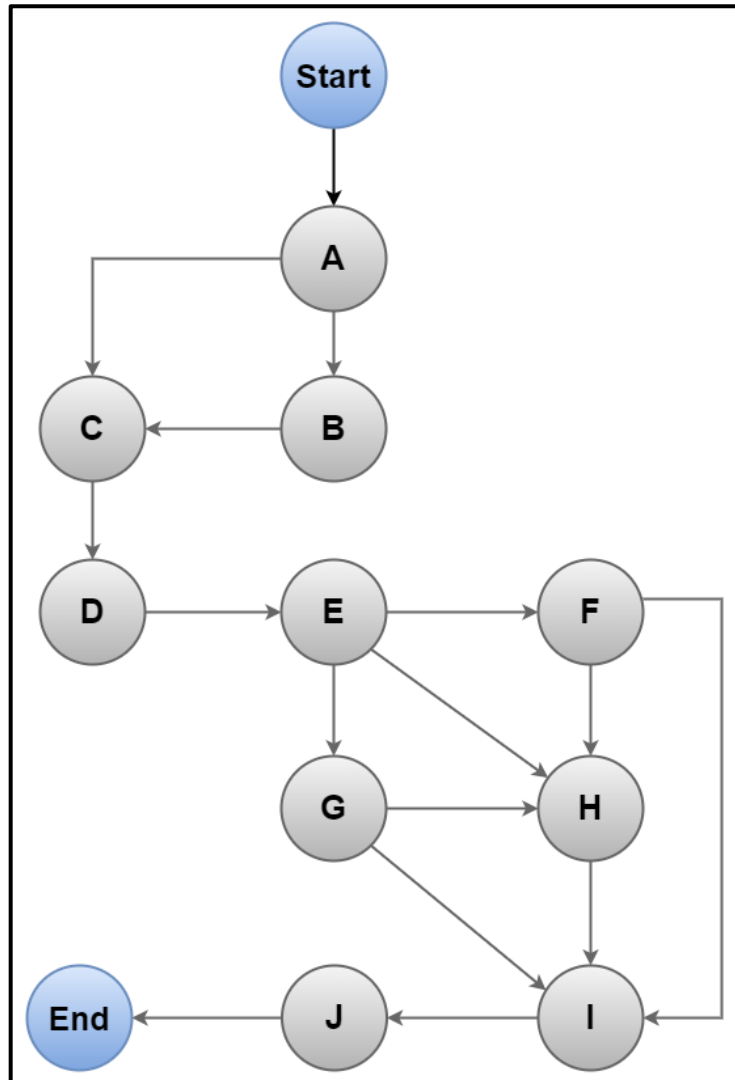


Figure 23: Network Diagram

13.4 Estimate Activity Completion Time

Activity ID	Dependency	Duration (days)
A	None	7
B	A	5
C	A, B	13
D	C	13
E	D	28
F	E	70
G	E	47
H	E, F, G	15
I	F, G, H	27
J	I	20

13.5 Identify the Critical Path (R7, n.d.)

Determining the following six parameters for each activity, which can identify the critical path:

ES: earliest start time: the earliest time at which the activity can start given that its precedent activities must be completed first.

$$ES(K) = \max [EF(J): J \text{ is an immediate predecessor of } K]$$

EF: earliest finish time: equal to the earliest start time for the activity plus the time required to complete the activity.

$$EF (K) = ES (K) + Dur (K)$$

LF: latest finish time: the latest time at which the activity can be completed without delaying the project.

$$LF (K) = \min [LS (J): J \text{ is a successor of } K]$$

LS: latest start time: equal to the latest finish time minus the time required to complete the activity.

$$LS (K) = LF (K) - Dur (K)$$

TS: Total Slack: the time that the completion of an activity can be delayed without delaying the end of the project

$$TS (K) = LS (K) - ES (K)$$

FS: Free Slack: the time that an activity can be delayed without delaying both the start of any succeeding activity and the end of the project.

$$FS (K) = \min [ES (J): J \text{ is successor of } K] - EF (K)$$

Activity	Duration	ES	EF	LS	LF	TS	FS
Start	0	0	0	0	0	0	0
A	7	0	7	0	7	0	0
B	5	7	12	7	12	0	0
C	13	12	25	12	25	0	0
D	13	25	38	25	38	0	0
E	28	38	66	38	66	0	0
F	70	66	136	66	136	0	0
G	47	66	113	104	151	38	23
H	15	136	151	136	151	0	0
I	27	151	178	151	178	0	0
J	20	178	198	178	198	0	0
End	0	198	198	198	198	0	0

The slack time for an activity is the time between its earliest and latest start time, or between its earliest and latest finish time. Slack is the amount of time that an activity can be delayed past its earliest start or earliest finish without delaying the project.

The critical path is the path through the project network in which none of the activities have slack, that is, the path for which $ES = LS$ and $EF = LF$ for all activities in the path. A delay in the critical path delays the project. Similarly, to accelerate the project it is necessary to reduce the total time required for the activities in the critical path.

13.6 Update CPM Diagram

As the project progresses, the actual task completion times will be known, and the network diagram can be updated to include this information. A new critical path may emerge, and structural changes may be made in the network if project requirements change.

The critical path is:

Start → A → B → C → D → E → F → H → I → J → End

14 VISION STATEMENT

Hotel Hype's aim is to be the leading platform that provides unbiased and accurate hotel recommendations by using the highest visibility and real-time location capabilities to provide travelers with confidence and personalized hotel selection. By combining user reviews, detailed information and recommendations, we aim to simplify the hotel selection process and improve the travel experience of the users all over the world and saves their time in this busy world.

15 RISK LIST

- **Electricity Load shedding:**

There is a huge problem of electricity shortages, just like in our country. The shortfall seems to be overcome these days, but its future status is still uncertain. If the load shedding starts again, the project may not be finish within certain deadlines.

- **Unrealizable/tight Timetable**

If the project dates are updated and the time period is condensed, the project may not be accomplished.

- **Resistance between Team and Management**

It can lead to failure if any of the jobs are not carried out in an appropriate manner and without adequate interest. In addition, if the opinion of the manager and team members are conflicting, the project is badly affected.

- **Procedural risks**

Day-to-day operational activities may be hamper by improper implementation of processes, conflicting priorities.

- **Requirements or gold-plating for developers:**

Gold plating is the process of meeting time-consuming excess requisites and can lead to project failures.

- **External Risks**

Our development team is not fully updated according to market trends and we do not have best business analyst in the available resources. There might be inevitable changes in customer product strategy and priority in near future.

16 REQUIREMENT TRACEABILITY MATRIX (R8)

The requirements trace-ability matrix is a table used to trace project life cycle activities and work products to the project requirements. The matrix establishes a thread that traces requirements from identification through implementation.

16.1 Visitor RTM

Sr. #	Para #	System Specification Text	Use Case Name	Category
1	1.0	A user “shall” register to the system.	UC_Registration	Online
2	1.0	The system “shall” provide the registration process to users.	UC_Registration	Online
3	1.0	The system “shall” send an OTP to the user's email for verification.	UC_EmailVerification	Online
4	1.0	The user “shall” enter the OTP to complete registration	UC_EmailVerification	Online
5	1.0	The user “shall” log in to the system after registration.	UC_Login	Online
6	1.0	System “shall” verify login credentials and provide access.	UC_Login	Online
7	1.0	The user “shall” browse and search for hotels based on preferences.	UC_BrowseSearch	Online
8	1.0	The user “shall” list a hotel and send a request to admin.	UC_ListHotel	Online
9	1.0	The user “shall” submit a review for existing hotels.	UC_ReviewSubmission	Online
10	1.0	The user “shall” search for hotels using live location.	UC_LiveLocationSearch	Online

16.2 Partner RTM

Sr. #	Para #	System Specification Text	Use Case Name	Category
1	1.0	A partner “shall” log in to the partner panel.	UC_Login	Online
2	1.0	The system “shall” provide access to the partner panel after login.	UC_AccessPartnerPanel	Online
4	1.0	The partner “shall” manage hotels by deleting a listed hotel	UC_deleteHotel	Online
5	1.0	The partner “shall” manage hotels by updating existing listed hotel	UC_UpdateHotels	Online
6	1.0	The partner “shall” update account details such as email, password, and username.	UC_UpdateAccountDetails	Online
7	1.0	The partner “shall” list a hotel and send a request to admin for approval.	UC_ListHotel	Online

16.3 Admin RTM

Sr. #	Para #	System Specification Text	Use Case Name	Category
1	1.0	An admin “shall” log in to the admin panel.	UC_Login	Online
2	1.0	The system “shall” provide access to the admin panel after login.	UC_AccessAdminPanel	Online
4	1.0	The admin “shall” manage user accounts by deleting them	UC_ManageUserAccounts	Online
5	1.0	The admin “shall” manage hotel by adding new hotels	UC_AddHotels	Online
		The admin “shall” manage hotel by deleteing existing hotels	UC_DeleteHotels	
6	1.0	The system “shall” send an email notification to the partner if their listing is deleted.	UC_EmailNotificationOnDeletion	Online
7	1.0	The admin “shall” view details of all hotels.	UC_ViewHotels	Online
8	1.0	The admin “shall” handle listing requests by approving or denying them.	UC_ReviewListingRequests	Online
9	1.0	The admin “shall” log out of the system.	UC_logout	Online

17 PROJECT/PRODUCT COSTING

17.1 Project Cost Estimation by COCOMO model

Key Metrics in COCOMO

- **Size Metric:** In COCOMO, the primary input is the estimated size of the project, typically measured in KLOC (thousands of lines of code)

Finally, Total Project Cost and Total Project Effort are calculated given below

$$\text{Effort} = a * (\text{KLOC})^b$$

Where:

- E = Effort (in person-months)
- KLOC = Estimated size of the software in thousands of lines of code (KLOC)
- a= A constant factor based on the project type
- b = An exponent factor based on the project type

17.1.1 Effort estimation

$$E = 2.4 * (7.9)^{1.05}$$

$$E \approx 2.4 \times 8.34$$

$$E \approx 20.03 \text{ person-month}$$

17.1.2 Development Time Estimation

$$T = c * (E)^d$$

$$T = 2.5 * (20.03)^{0.38}$$

$$T \approx 2.5 * 4.84$$

$$T \approx 12.1 \text{ months}$$

17.1.3 Total Cost Estimation

$$C_{\text{total}} = E * \text{Cost per person-month}$$

(assuming Cost per person-month = 10000Rs)

$$C_{\text{total}} = 20.03 * 10,000$$

$$C_{\text{total}} = 200,300\text{Rs}$$

18 SCREEN IMAGES

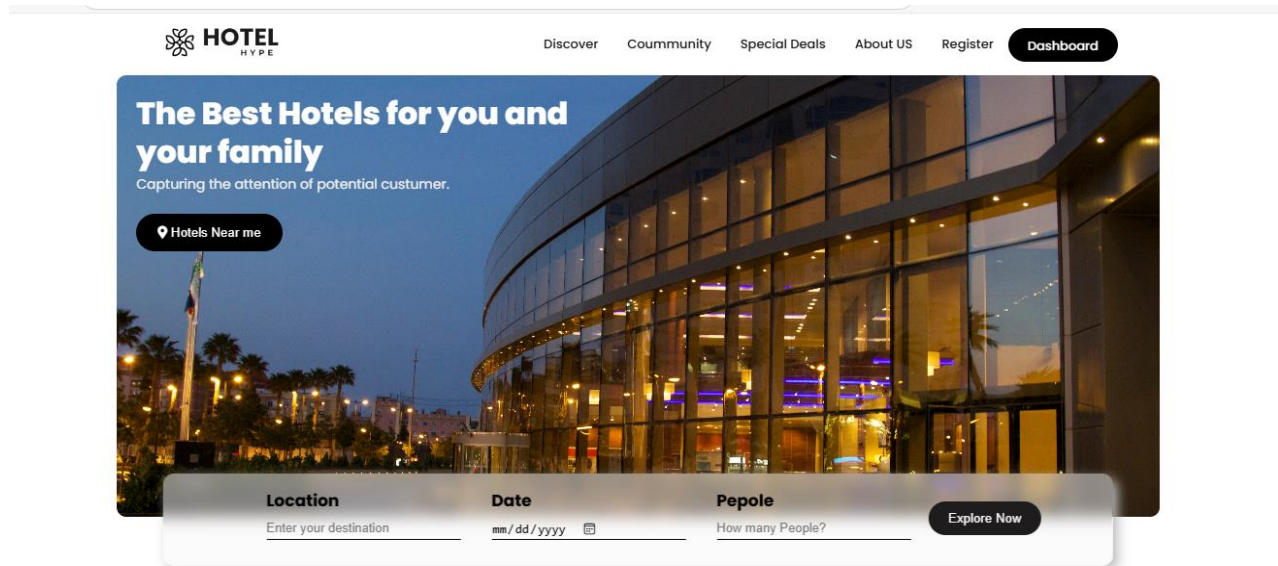


Figure 24: Home Page

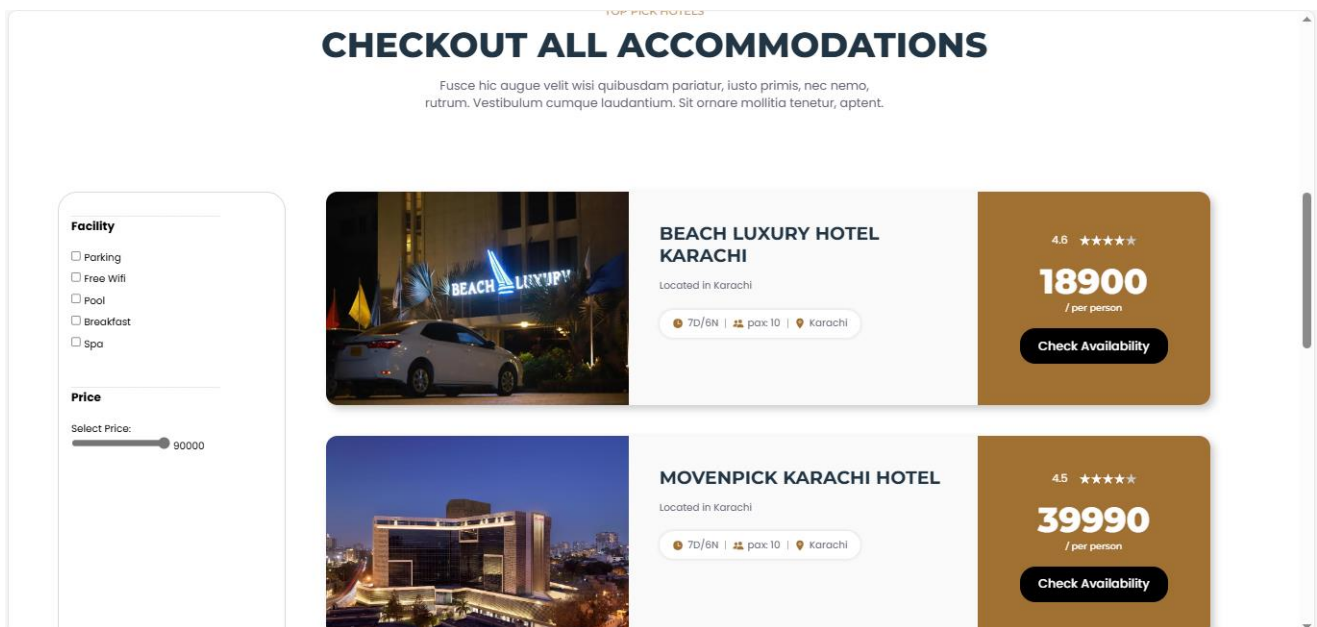


Figure 25: Hotels Page

Guest Reviews

4.6

15,479 reviews

[Read all reviews](#)

See Availability

Facilities

Room

7.5

Facilities

8.6

Comfort

↓

1

Staff

7.4

Cleanliness

7.6

Guests who stayed here loved

A

Ali Khan

Rooms are perfectly cleaned. Muzammil is a very nice and humble guy who can keep your rooms as new and...

S

sheraz mian

The beach luxury hotel in Karachi offers a delightful blend of indulgence and seaside serenity. Nestled along the shores of...

W

Warda Ahmed

Was a nice hotel. Wouldn't say amazing but the rooms were great! Of course it's by the water so it's...

N

Nisha Kumari

We had an event stakeholders and officials and arranged best bu

Figure 28: Hotel Detail Page (3)

HOTEL HYPE

Register Now

Username

Email address

Password

Confirm Password

Create Account

Already have an account!

Login Now

By signing in or creating an account, you agree with our [Terms & conditions](#) and [Privacy statement](#).

Figure 29: Registration Page

Govt. Graduate College, Civil Lines Sheikhpura

Page |



Login Now

Username

Email address

Password

Login

Already have not account

Create Account

By signing in or creating an account, you agree with our [Terms & conditions](#) and [Privacy statement](#).

All rights reserved.
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Figure 30: Login Page



Register Now

Enter OTP

Please enter the 6-digit OTP sent to your email.

2

3

4

5

6

7

Verify

Already have an account!

Login Now

By signing in or creating an account, you agree with our [Terms & conditions](#) and [Privacy statement](#).

Figure 31: OTP Verification


HOTEL HYPE

Hotel Hype Listing Form!




Get yourself Registered!

Name

Address

Phone Number

Back
Next



WHAT NEEDS TO BE INCLUDE?

- A Complete name, email, password (Password must be strong)
- Include both your street name and number for the entire property
- The floor number in address line 2 where relevant
- Individual apartment or floor numbers can be shared later
- Provide the post/zip code
- Correctly spell the street name
- Use the physical address of the property, not your office or home address

WHAT FACILITY TO BE INCLUDE?

- The facilities listed here are the ones most searched for by guests
- After you complete your registration, you can add more facilities from a larger list in the extranet, the platform you'll use to manage your property.

WHAT IF MY HOUSE RULES CHANGE?

- You can easily customise these house rules later and additional house rules can be set on the Policies page of the extranet after you complete registration.

Figure 32: Hotel Listing Form Page


HOTEL HYPE

Admin Panel

Logout

Dashboard
User Management
Hotels Management
Approvals and Requests
Content Management

Dashboard



Listed Hotels
23



Pending Requests
1



Recent Reviews
25



Total Registrations
9

Pending Requests

Hotel Owner	Hotel Name	Email Address	Status
	15	abeera@gmail.com	pending

Figure 33: Admin Panel

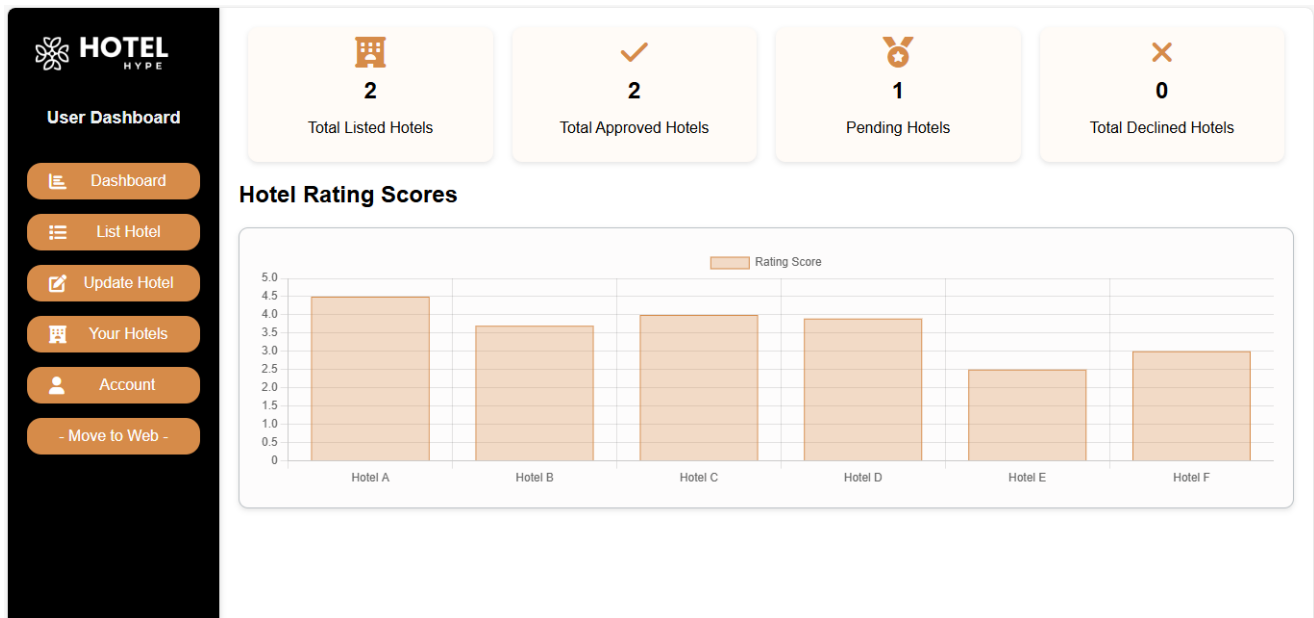


Figure 34: Partner Panel

19 CONCLUSION

Hotel Hype is a new solution for the hotel industry that solves the fundamental problem of transparency in hotel selection and evaluation.

Combining real services, thoughtful reviews and powerful reviews, the site offers users reliable, unbiased, personal recommendations. The platform allows users to make informed decisions with accurate hotel reviews and general information. In addition, Hotel Hype supports social interaction by allowing users to contribute their own reviews, allowing Hype Partners to maintain high-quality hotel listings. Focused on user satisfaction and data-driven insights, Hotel Hype is poised to revolutionize the hotel booking and recommendation space by providing travelers with good information and confidence.

20 FUTURE WORK

Looking Ahead Hotel Hype aims to improve its quality by integrating content-based recommendation system. This feature will use advanced algorithms to analyze user data, preferences, and review feedback to create personalized hotel recommendations. By customizing recommendations based on past user interactions and feedback, this development will increase the effectiveness and satisfaction of the recommendations provided. Additionally, the integration of the booking function directly into the website will enhance the user experience, allowing users to explore and make the hotels recommended for booking. This booking feature will make the job easier for users and will provide a solution for the entire process from hotel selection to booking confirmation. As part of future development, Hotel Hype will also focus on improving user information, allowing users to specify detailed information and preferences, and the recommendation model will be used to improve and refine the recommendation over time. The platform is also exploring the development of mobile applications to provide easier and more convenient access to all operating platforms, including recommendations and booking activities. Collaboration with hotels and travel agencies to provide special promotions and promotions, while ensuring the availability of new information and reservations, is another important point. Finally, as the platform develops, Hotel Hype will prioritize the use of data security and privacy to protect user data, especially when making reservations and payments. By focusing on these areas, Hotel Hype aims to develop into a mature platform that not only provides hotel recommendations, but also facilitates the travel planning and booking process.

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